5. Airport Operations



5. Airport Operations

5.1 Airport Elements and Operations

5.2 Airport Planning

5.3 Airport Capacity



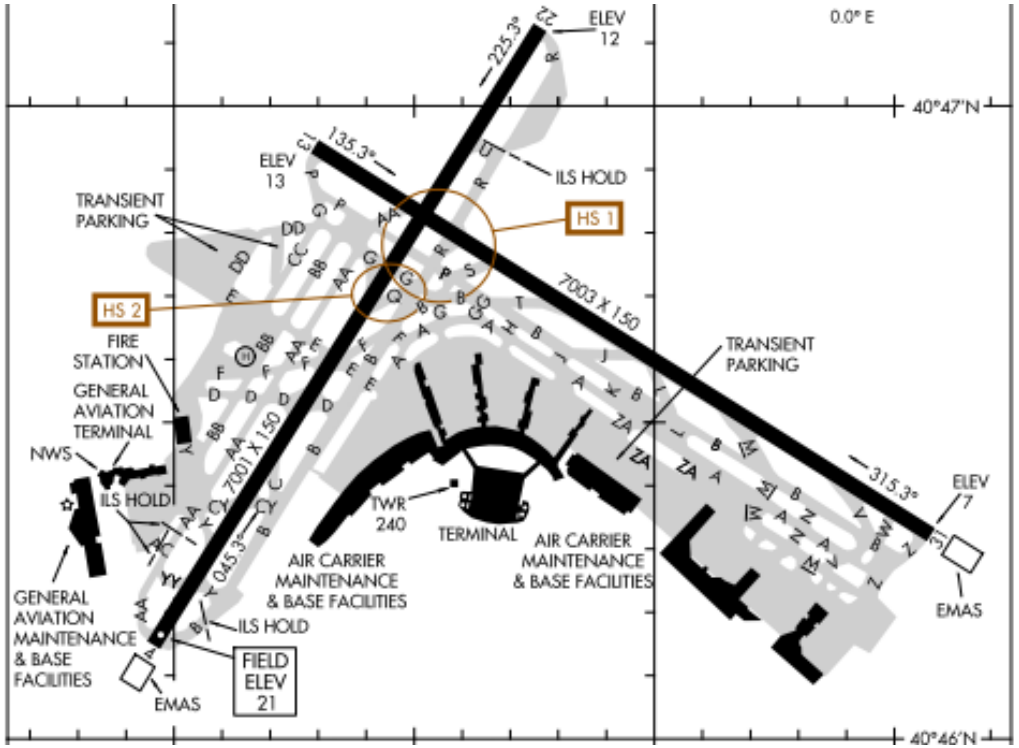
Source: Munich Airport

5. Airport Operations

5.1 Airport Elements and Operations

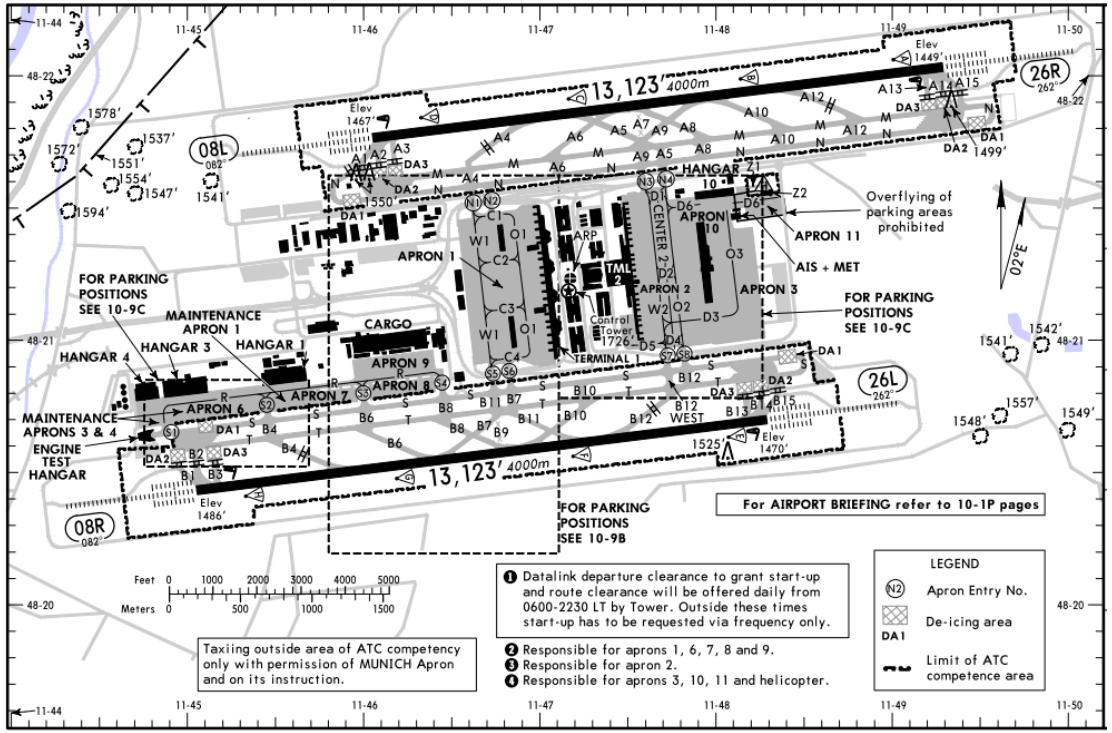


Example 1: New York La Guardia



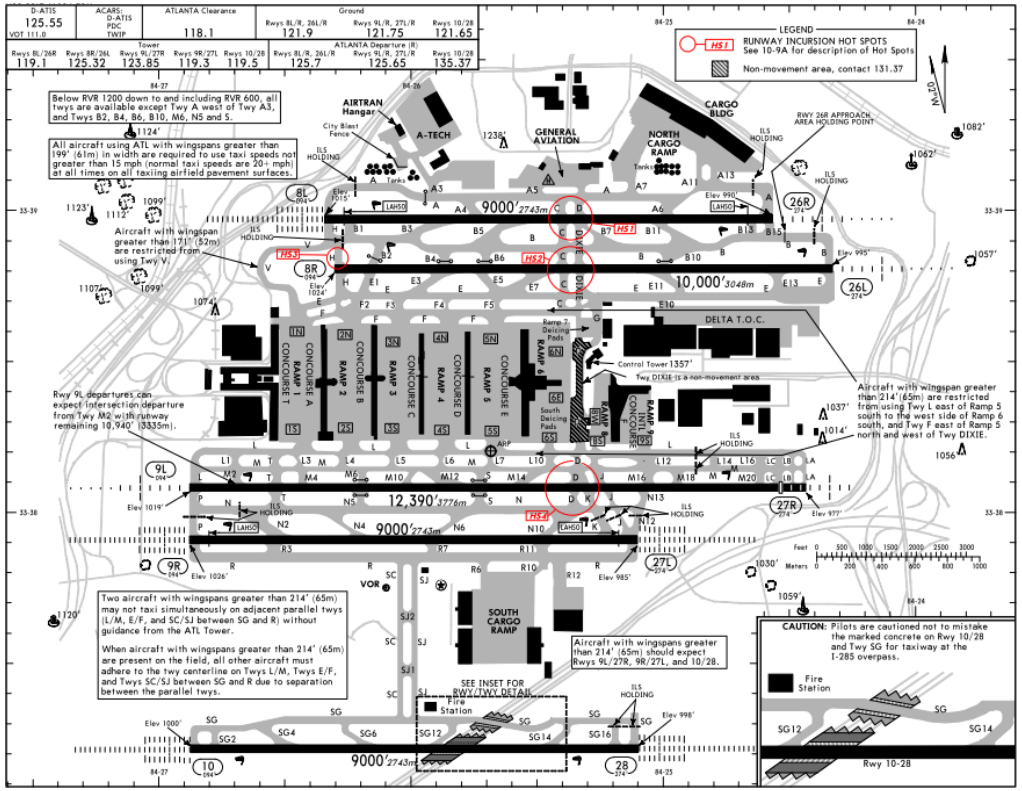
- Two intersecting runways
- Normally operated with two active runways, one used primarily for arrivals and the other for departures
- In strong wind / poor weather conditions, all movements may be accommodated on a single runway

Example 2: Munich



- Two independent parallel runways
- Northern runway used primarily for north-bound departures and arrivals, southern runways for south-bound traffic
- Two types of passenger terminals: T1 for O-D traffic (point-to-point network), T2 for connecting traffic (hub-and-spoke network)

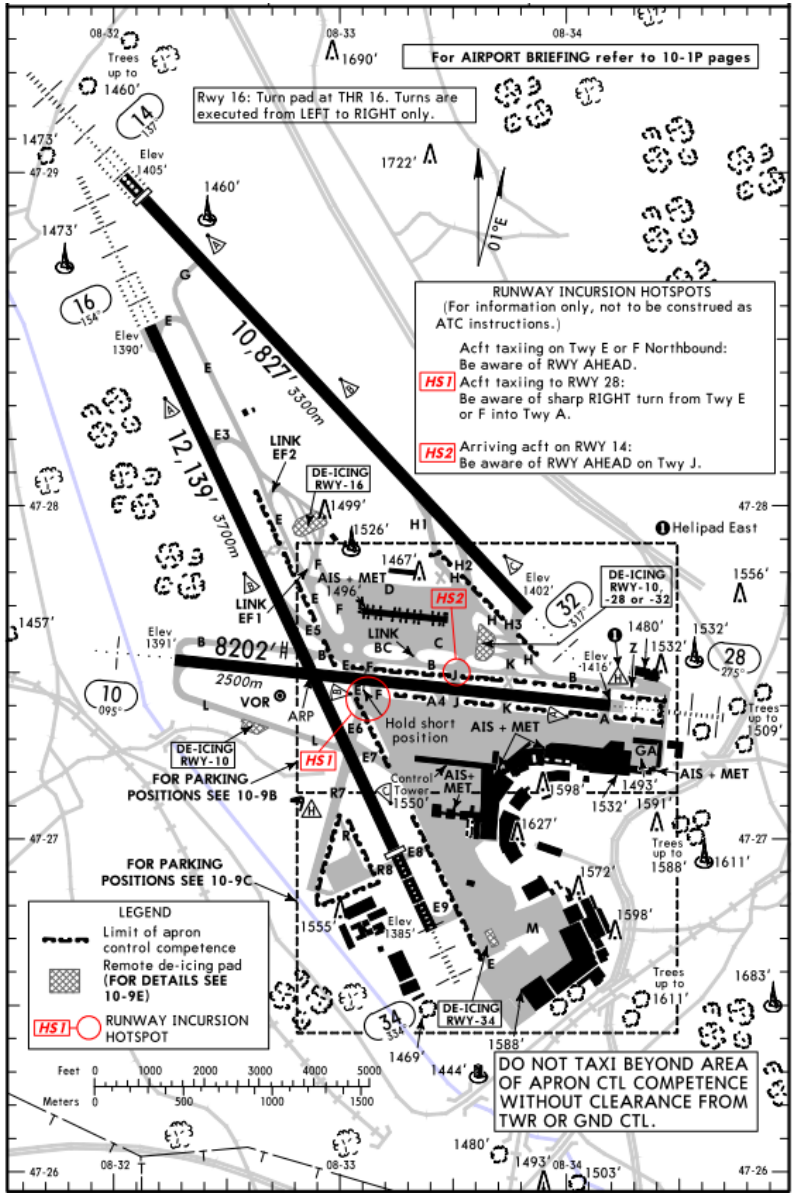
Example 3: Atlanta International



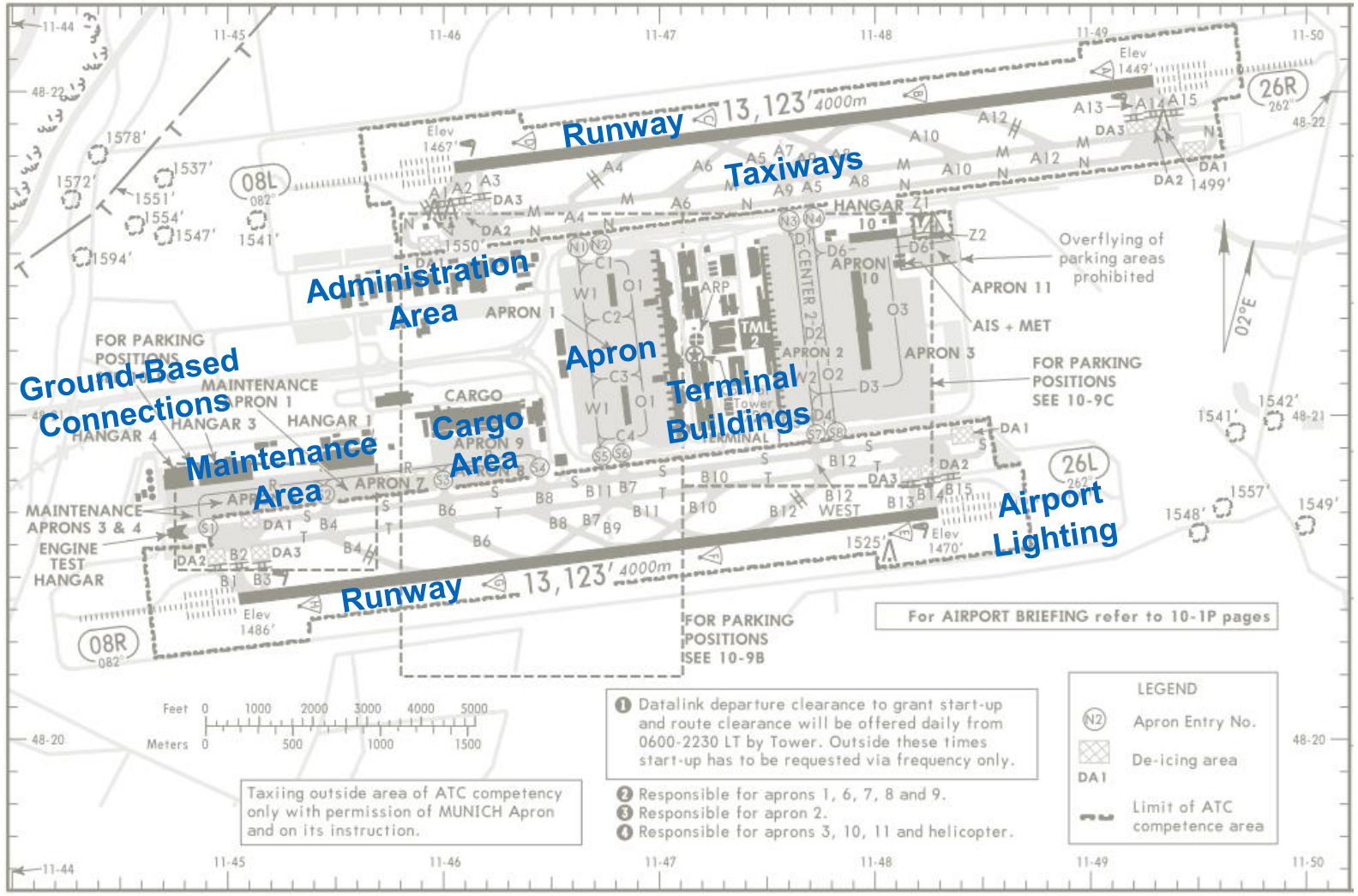
- Main runway system consists of two pairs of close parallel runways
- Typically, runways 08L/26R and 09R/27L are used for arrivals and the other for departures
- Passenger terminal consisting of a main terminal and five linear mid-field satellites, all connected via an underground passage and automated people mover

Example 4: Zurich

- Three runways:
  - Runway 14 primarily used for landing
  - Runway 28 primarily used for light and medium departures
  - Runway 16 primarily used for heavy departures (MTOW > 136 tons)
- Runway system restricts traffic handling capacity
- Two passenger terminals at southern area of the airport, “Dock Midfield” positioned in the middle of the airport area especially for Non-Schengen traffic

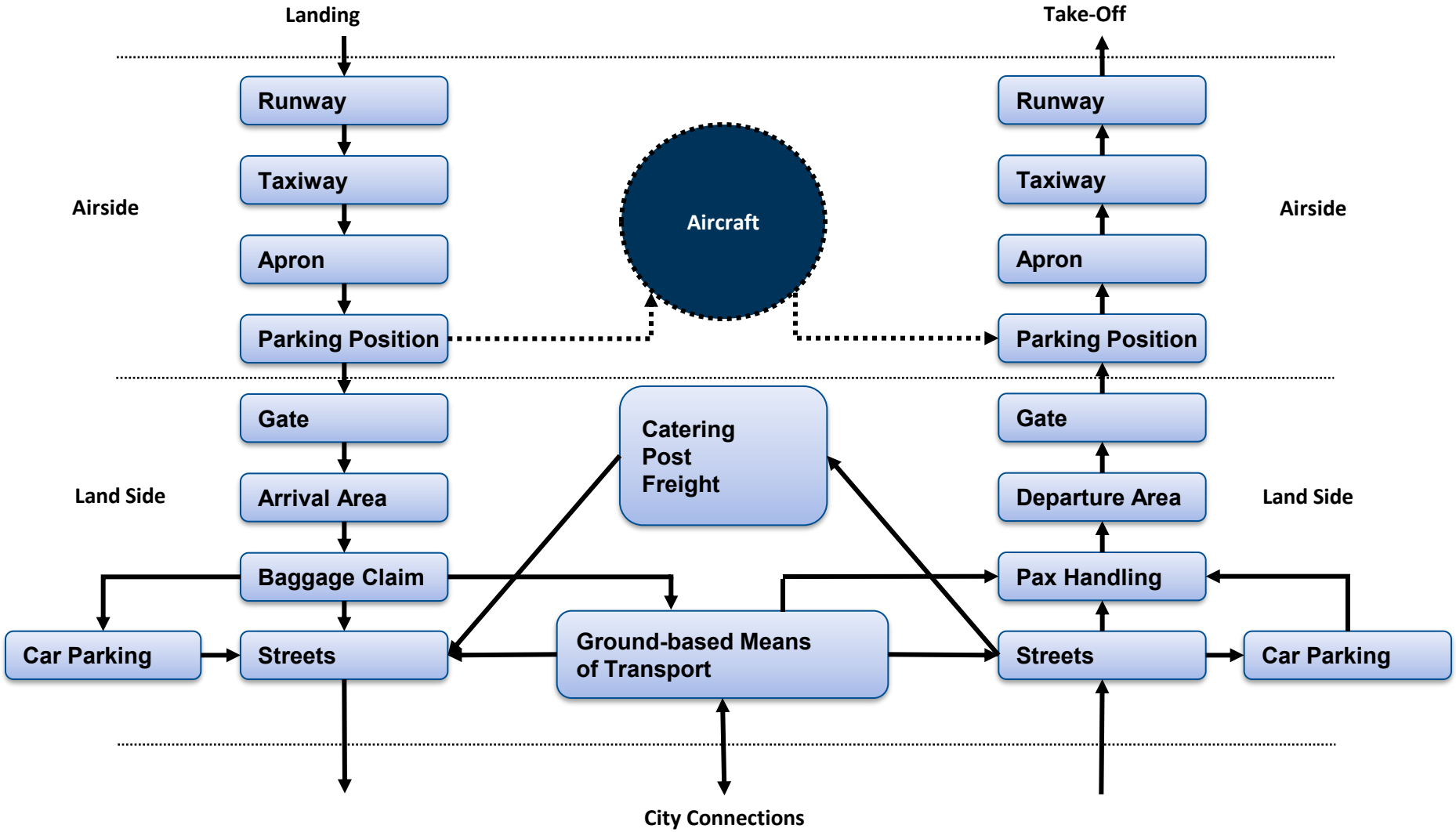


Airport Layout (Munich Airport)

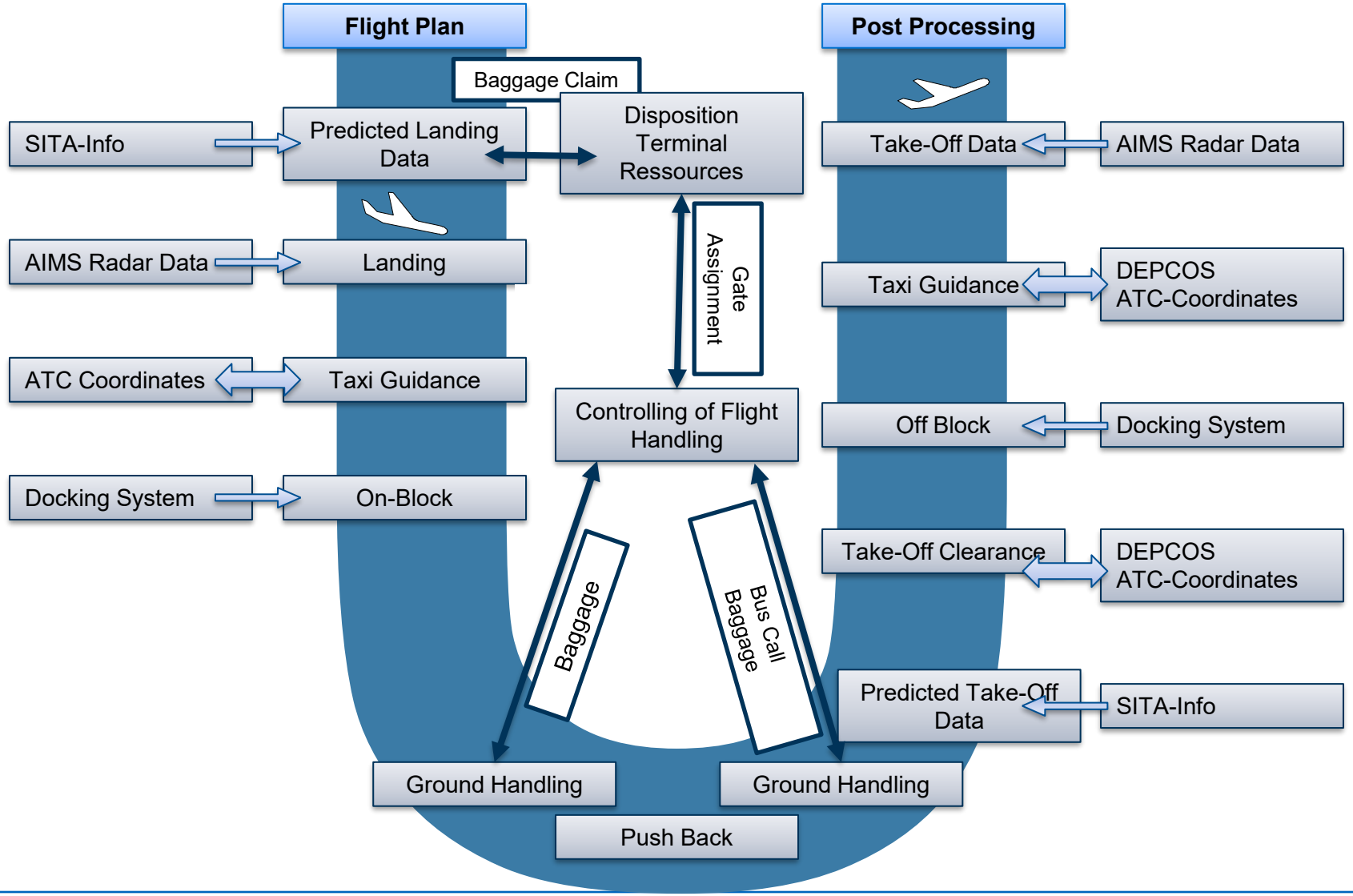


Source: Jeppesen

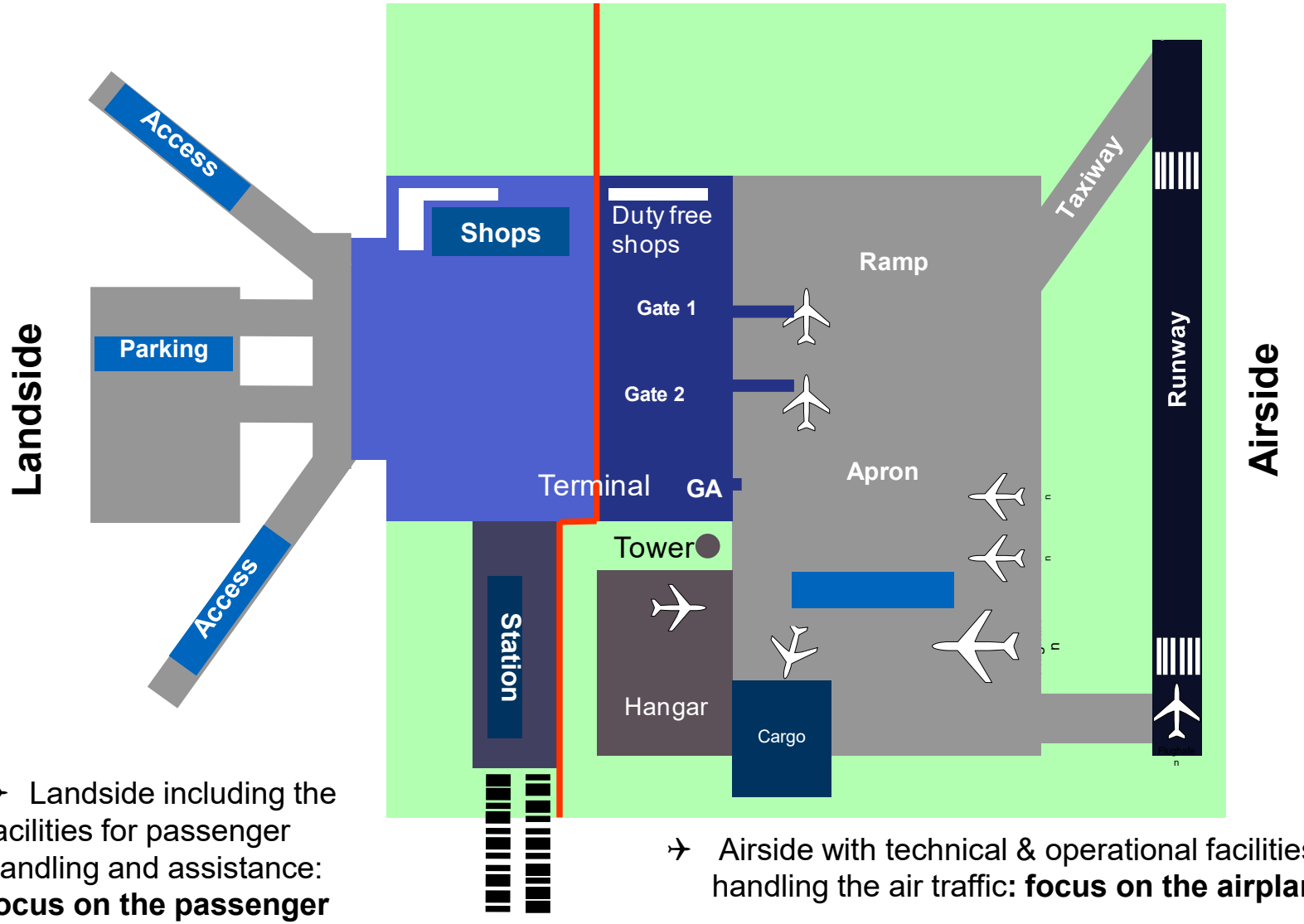
The Airport System



Airside Operational Process Chain



Airport Elements

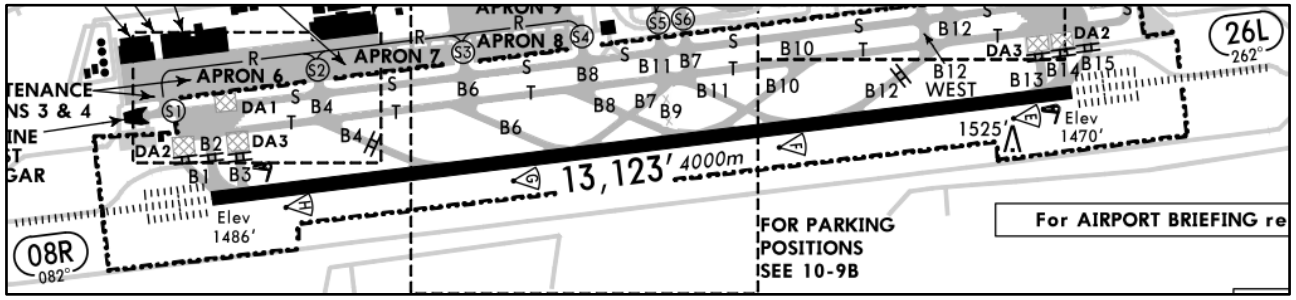


✈ Landside including the facilities for passenger handling and assistance: **focus on the passenger**

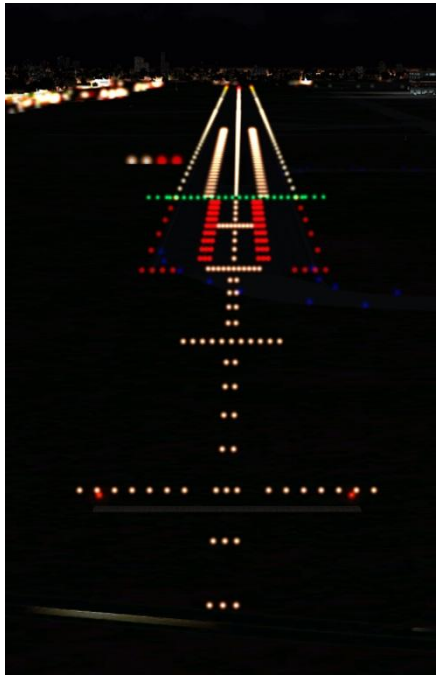
✈ Airside with technical & operational facilities for handling the air traffic**: focus on the airplane**

Airside Elements of the Airport (1)

Runway

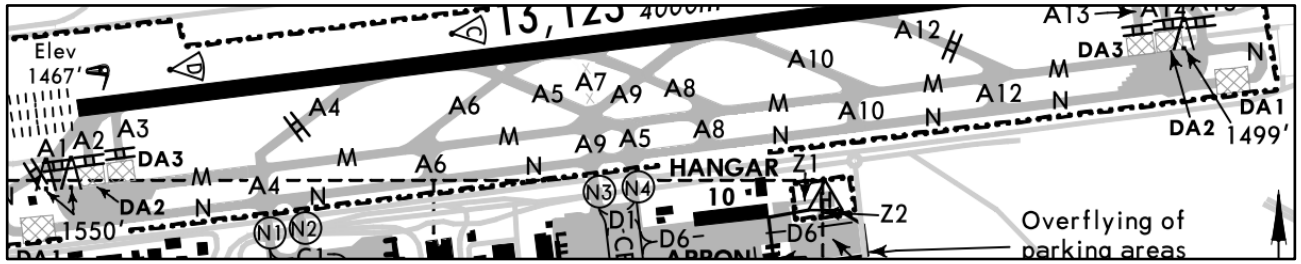


- Runway
  - made of concrete or bituminous construction
  - dimensioning in compliance with ICAO guidelines and mean sea level
- Navigation lighting
  - Border lighting
  - Middle line lighting
  - Edge lighting
  - Approach lighting
- Signage

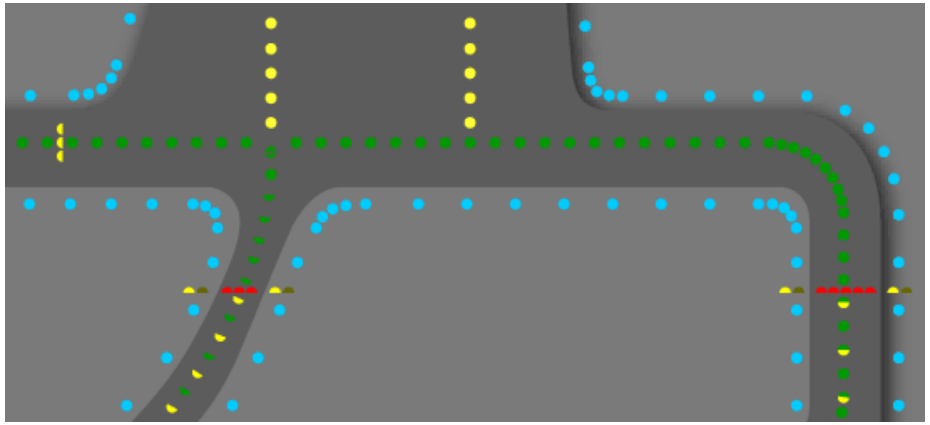


Airside Elements of the Airport (2)

Taxiways



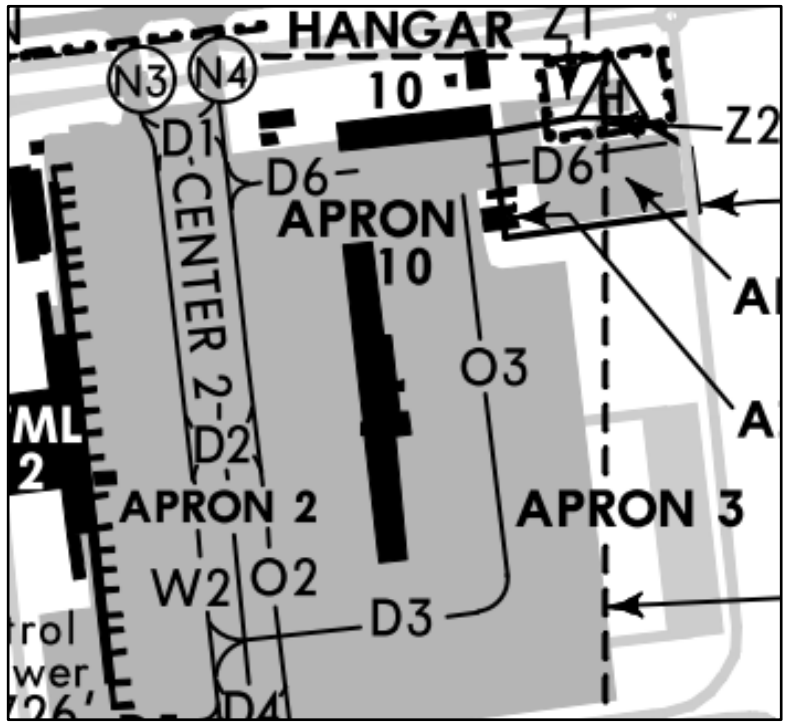
- Taxiway
  - made of concrete or bituminous construction
  - dimensioning in compliance with ICAO guidelines
- Navigation lighting
  - Border lighting
  - Middle line lighting
  - Stop Bar lighting



- Signage
- Taxi guidance surveillance systems, e.g. induction loops

Airside Elements of the Airport (3)

- Apron
  - mostly made of concrete
  - rarely bituminous construction
- Air Traffic Control Facilities
  - Tower
  - Ground control
- Navigation aids
  - Taxi guidance systems
  - Runway & apron lighting
- Technical facilities
  - Fuel tanks
  - Maintenance hangar
- Docking systems
- Boarding bridges
- Parking positions
- PAX & Cargo Terminals
- Areas for parking operational devices



Optional:

- Electric power supply
- Fuel supply (hydrants)
- Pre-Conditioned Air Units (PCA)
- Air Starter Units (ASU)

Airside Elements of the Airport (4)

Passenger Handling (Terminal)

Primary Elements

- Check-In Counters
- Ticket Counters
- Security Control
- Waiting Areas
- Baggage Systems
- Customs Control
- Immigration Control
- Border Control
- Baggage Claim
- Medical Services

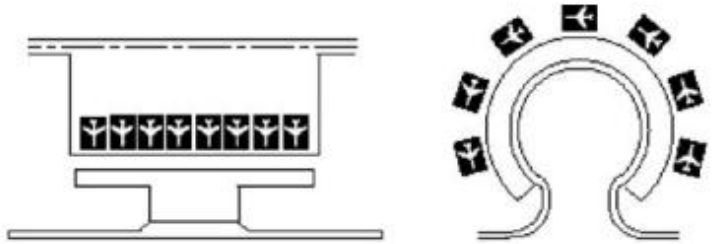
Secondary Elements

- Lost & Found
- Restaurants & Catering Services
- Travel Agencies
- Spiritual Welfare
- Travel Requisites and other commercial facilities
- Lounges (Business / First Class)
- Tourist Information Center

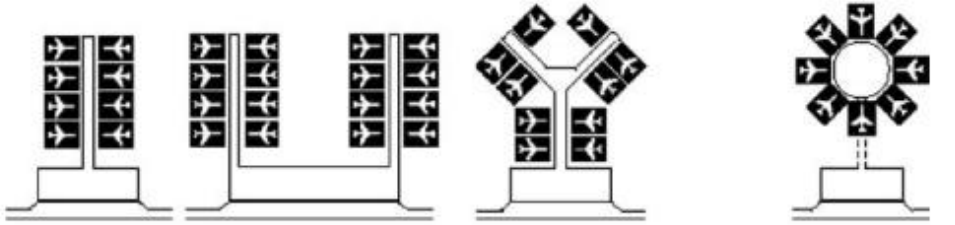


Airside Elements of the Airport (5)

Passenger Terminals

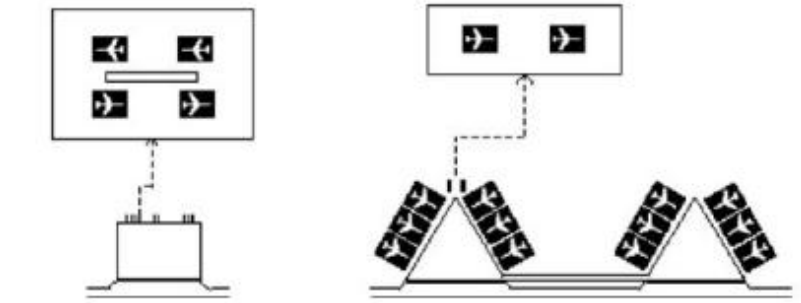


Linear concept and its variations



Pier (finger) concept

Satellite concept



Transporter (open apron) concept

Hybrid concept

Source: ICAO (1999)

Other Major Airport Facilities



- Car Parking Areas
- Street Connections
- Public Transport Connections
- Airport Maintenance and Winter Services
- Wastewater Facilities
- Security Fences and Gates
- Hotels
- Industry Facilities
- Safety Facilities (fire department, medical services, etc.)





5. Airport Operations

5.2 Airport Planning



General Airport Requirements and Planning Factors

Requirements

- Safe operations
- 24/7-operability
- Economic efficiency
- Intermodal connections
- Minimization of environmental impact
- Acceptance with local population

Planning Factors

- Number and direction of runways
- Number of taxiways
- Size and shape of apron
- Terrain conditions of available land surface
- Navigation obstacles
- Land surface available for future expansion projects
- Meteorological conditions
- Size of planned airport facilities

Airside Planning Fundamentals

ICAO Regulations

Contract basis:

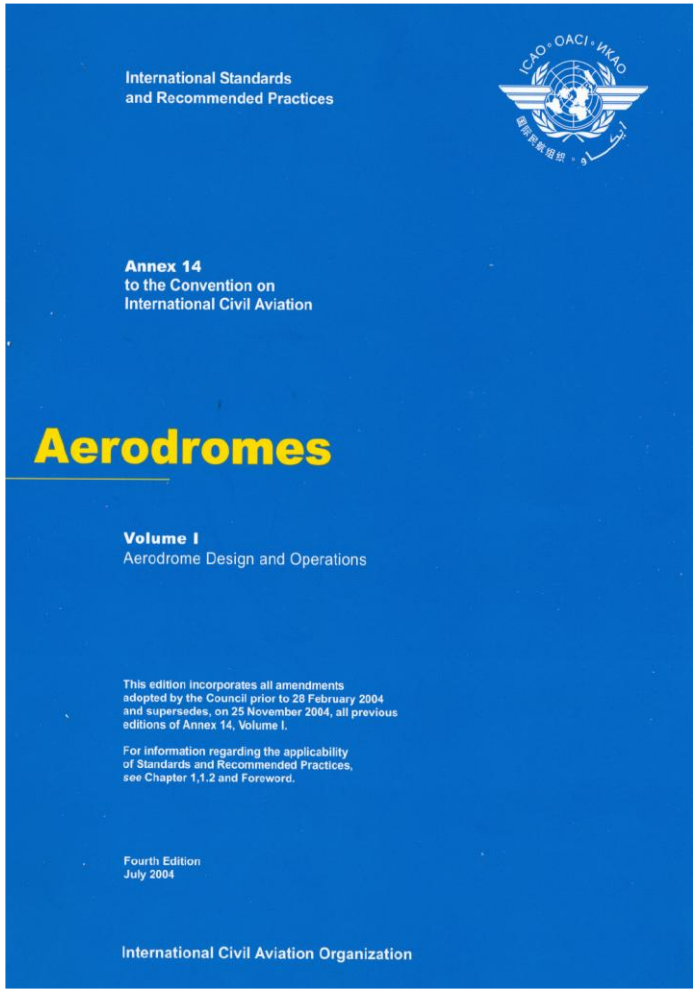
Convention on International Civil Aviation of 7 December 1944 (Chicago) and their annexes (Annex 1 - 18) and the Manuals

- ICAO, acting on the basis of convention specifications for the design, construction, and operation of airports and their facilities. Goal: the establishment of international **quality** and **safety standards**.
- The ICAO fundamentally distinguishes between “**Standards**" and "**Recommended Practices**."
- **Standards** are compulsory: minimum planning requirements for all airports of the member states.
- **Recommended Practices** are non-compulsory recommended implementation guidelines.

ICAO Guidelines for Airport Planning (1)

ICAO Annexes 1-18

Annex	Annex
1. Personnel Licensing	**11.** Air Traffic Control,- Flight Information,- and Alerting Service
2. Rules of the Air	**12.** Search and Rescue
3. Meteorological Service for International Air Navigation	**13.** Aircraft Accident and Incident Investigation
4. Aeronautical Charts	**14. Aerodromes** **a) Aerodrome Design and Operations** **b) Heliports**
5. Units of Measurement to be used in Air and Ground Operations.	**15.** Aeronautical Information Services
6. Operation of International Aircraft a) Commercial Air Transport Aeroplanes b) General Aviation - Aeroplanes c) Operations - Helicopters	**16.** Environmental Protection a) Aircraft Noise b) Aircraft Engine Emissions
7. Aircraft Nationality and Registration	**17.** Security - Safeguarding Int. Civil Aviation against Acts of Unlaw
8. Airworthiness of Aircraft	**18.** The Safe Transport of Dangerous Goods by Air
9. Facilitation	
10. Aeronautical Telecommunications	



ICAO Guidelines for Airport Planning (2)

- Annex 14 -

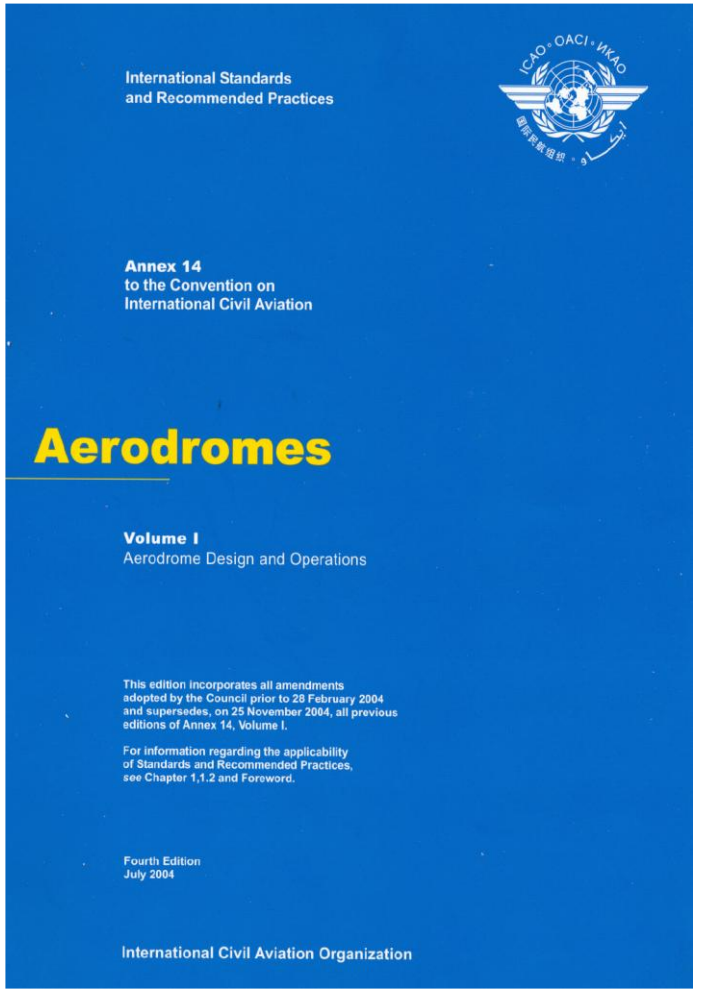
PART I: AERODROME DESIGN AND OPERATIONS

a) Aerodrome Data

- In order to classify airports, an **Aerodrome reference code (ARC)** is defined

- The classes are related to **aircraft categories (aircraft reference code)**. As a result, various qualitative and quantitative planning projects for the airports can be determined.

- The Aerodrome Reference Code is composed of:
  - **Aeroplane reference field length**
  - **Aircraft reference code** consisting of
    - the aircraft wingspan
    - the wheelbase of the outer main gear



Airside Design Standards

ICAO Aerodrome Reference Code (ARC)

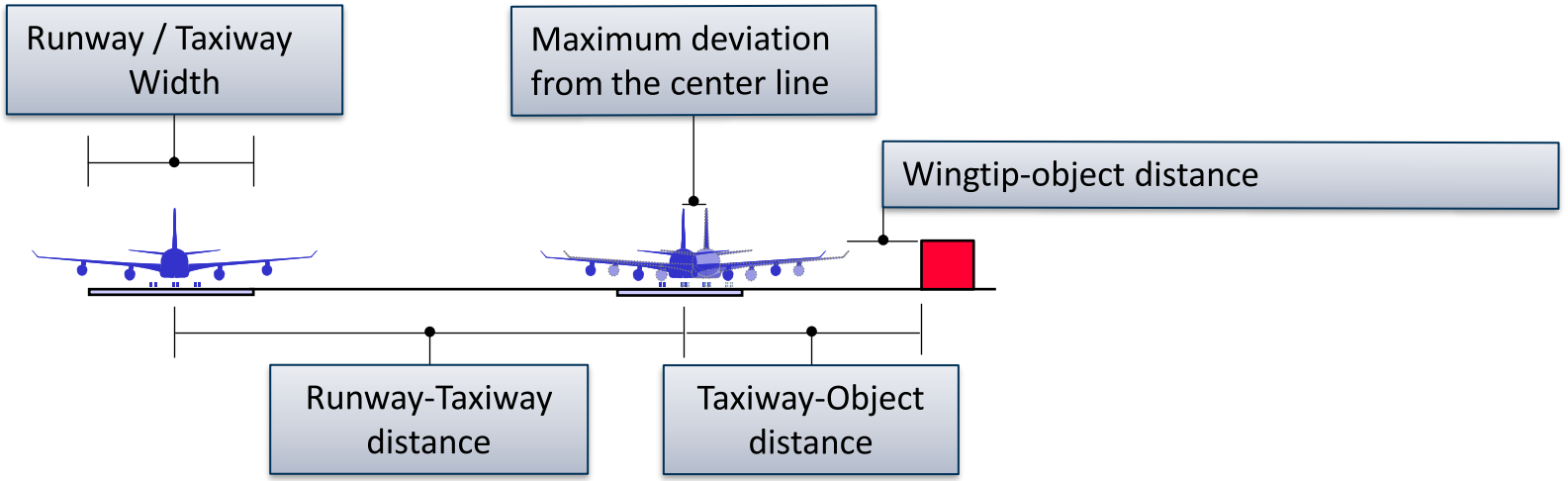
CODE ELEMENT 1		CODE ELEMENT 2		
Code number	Aeroplane reference field length	Code letter	Wing span	Outer main gear wheel span[a]
(1)	(2)	(3)	(4)	(5)
1	Less than 800 m	A	Up to but not including 15 m	Up to but not including 4.5 m
2	800 m up to but not including 1 200 m	B	15 m up to but not including 24 m	4.5 m up to but not including 6 m
3	1 200 m up to but not including 1 800 m	C	24 m up to but not including 36 m	6 m up to but not including 9 m
4	1 800 m and over	D	36 m up to but not including 52 m	9 m up to but not including 14 m
		E	52 m up to but not including 65 m	9 m up to but not including 14 m
		F	65 m up to but not including 80 m	14 m up to but not including 16 m

a. Distance between the outside edges of the main gear wheels.

- Classification of aircraft in Airport Reference codes for airport planning
- 2 code elements: flight performance + geometric properties

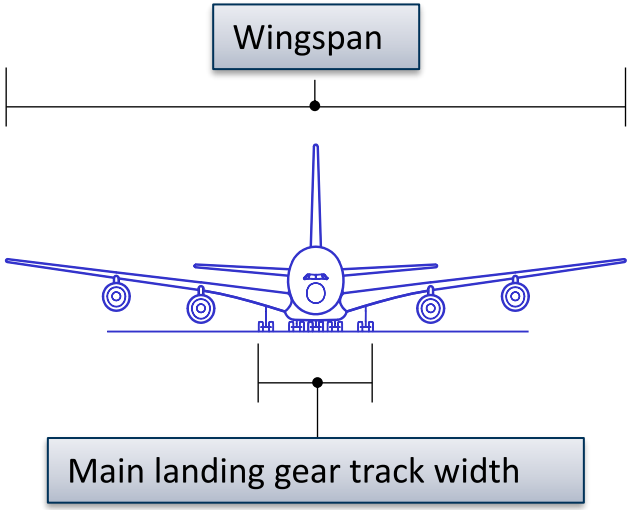
Source: ICAO

Aircraft Dimensions and References



6 Airport Codes
(Definition of geometric minimum requirements)

Airplane parameters



ICAO Guidelines for Airport Planning (3)

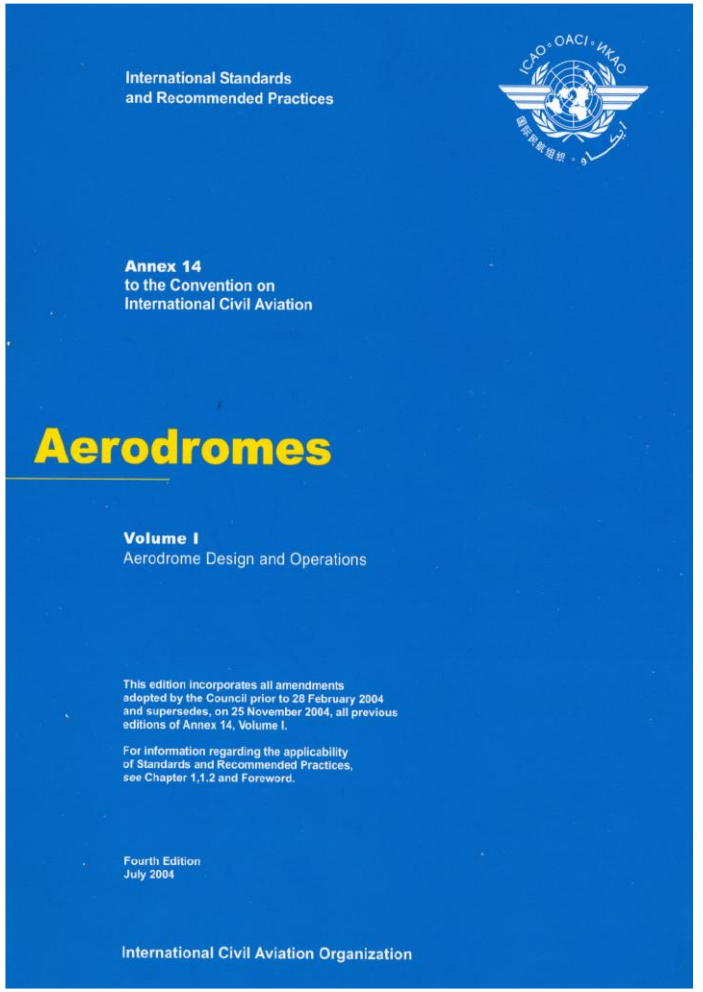
- Annex 14 -

PART I: AERODROME DESIGN AND OPERATIONS

b) Physical Characteristics

This chapter describes (among others) the requirements to topography and design of the runway system, the apron, the aircraft parking positions, and the taxiways with respect to:

- Dimensioning (length, width, height)
- Admissible lengthwise and crosswise gradients
- Curve radii
- Constructional and qualitative surface design (profiles)
- As a function of the airport reference code, allowances of safety clearances between the functional areas are defined in particular



ICAO Guidelines for Airport Planning (4)

- Annex 14 -

PART I: AERODROME DESIGN AND OPERATIONS

c) Obstacle restriction and removal

When building an airport, certain obstacle clearances are to be maintained for safe airport operations. Buildings and other artificial and natural obstacles of the environment of the airport are only permitted to have certain limited heights.

d) Equipment and Installations

Among others, the installation and the operation of technical equipment (e.g., electric energy supply, safety lighting, etc.) are defined here.

e) Emergency and other Services

The requirements to rescue services, airport fire department, and emergency planning are defined here.

2.2 PART II: Heliports

The planning prescriptions of Part I (Aerodrome Design and Operations) are analogously valid for helicopter airfields, however aircraft-specific requirements are adapted.



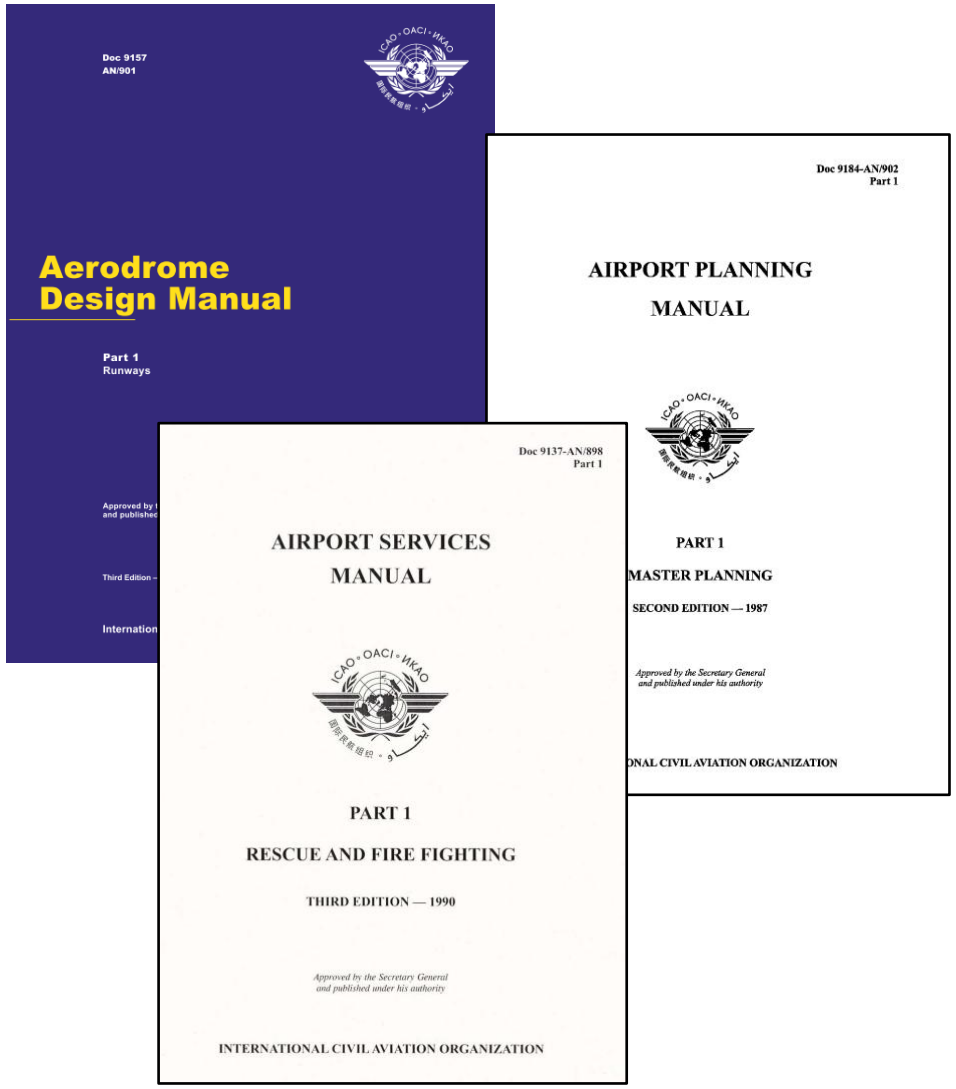
ICAO Guidelines for Airport Planning (5)

- Annex 14 -
MANUALS

Manuals are added to the ICAO convention and its annexes. The books based on Annex 14 specify the basic planning prescriptions at a high level of detail.

As far as Annex 14 is concerned, the following major manuals are:

- Aerodrome Design Manual
- Airport Planning Manual
- Airport Service Manual
- Heliport Manual



Airside Design Standards

FAA Airport Reference Code

FAA Reference code element 1		FAA Reference code element 2		
Aircraft approach category	Aircraft approach spped (AS) in knots	Airplane design group	Aircraft wing span (WS) in feet (m)	Tail height (TH) in feet
A	AS < 91	I	WS < 49 ft (15 m)	WS < 20
B	91 ≤ AS < 121	II	49 ft (15 m) ≤ WS < 79 ft (24 m)	20 ≤ WS < 30
C	121 ≤ AS < 141	III	79 ft (24 m) ≤ WS < 118 ft (36 m)	30 ≤ WS < 45
D	141 ≤ AS < 166	IV	118 ft (36 m) ≤ WS < 171 ft (52 m)	45 ≤ WS < 60
E	166 ≤ AS	V	171 ft (52 m) ≤ WS < 214 ft (65 m)	60 ≤ WS < 66
		VI	214 ft (65 m) ≤ WS < 262 ft (80 m)	66 ≤ WS < 80

Source: FAA

Aircraft Design – Aerodrome Reference Codes

Airbus

Aircraft Types

(ICAO)

Aircraft model	ICAO Aerodrome Reference Code[1]
A318	3C
A318 Sharklet*	3C
A319	4C
A319 Sharklet*	4C
A320	4C
A320 Sharklet*	4C
A321	4C
A321 Sharklet*	4C
A300	4D
A300-600	4D
A310	4D
A330-200	4E
A330-200F	4E
A330-300	4E
A340-200	4E
A340-300	4E
A340-500	4E
A340-600	4E
A350-900*	4E
A380-800	4F

Boeing

Aircraft Types

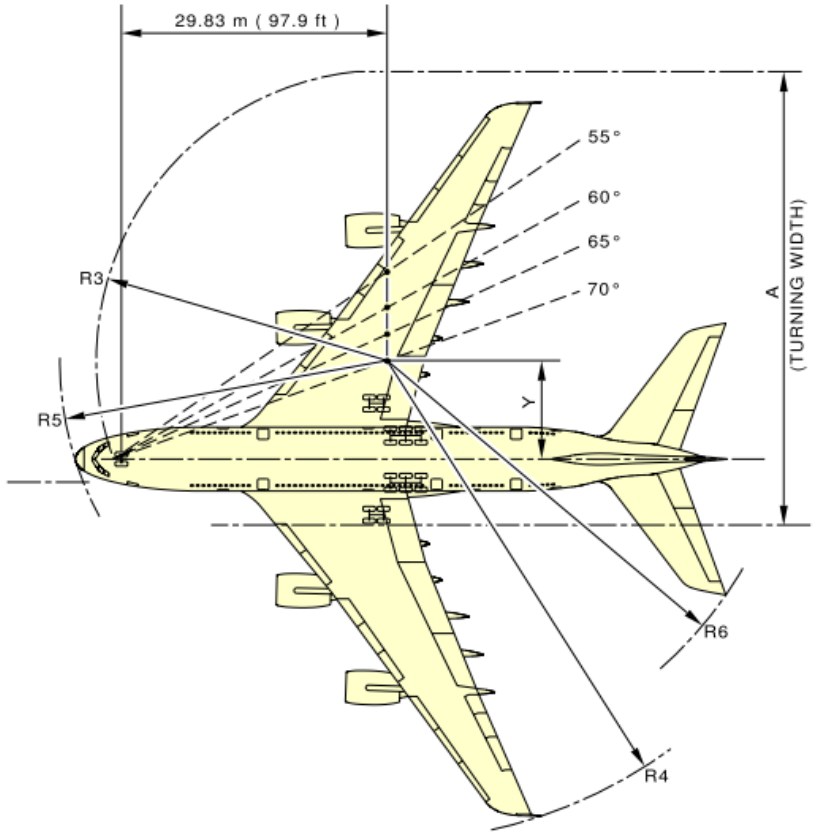
(FAA)

Boeing 737-300	C-III
Boeing 737-400	C-III
Boeing 737-500	C-III
Boeing 737-600	C-III
Boeing 737-700	C-III
Boeing 737-700W	C-III
Boeing 737-800	D-III
Boeing 737-800W	D-III
Boeing 737-900	D-III
Boeing 737-900W	D-III
Boeing 737-900ER	D-III
Boeing 737-900ERW	C-III
Boeing BBJ	C-III
Boeing BBJ 2	D-III
Boeing BBJ 3	C-III
Boeing 747-100	D-V
Boeing 747-200	D-V
Boeing 747-200F	D-V
Boeing 747-300	D-V
Boeing 747-400	D-V
Boeing 747-400F	D-V
Boeing 747-400ER	D-V
Boeing 747-SP	C-V
Boeing 747-8I*	D-VI
Boeing 747-8F*	D-VI
Boeing 757-200	C-IV
Boeing 757-300	D-IV
Boeing 767-200	C-IV
Boeing 767-200ER	D-IV
Boeing 767-300	C-IV
Boeing 767-300ER	D-IV
Boeing 767-400ER	D-IV
Boeing 777-200	C-V

Source: Airbus, Boeing

Example: Airbus A380 (1)

Turning Radii



A380–800/800F Minimum Turning Radius

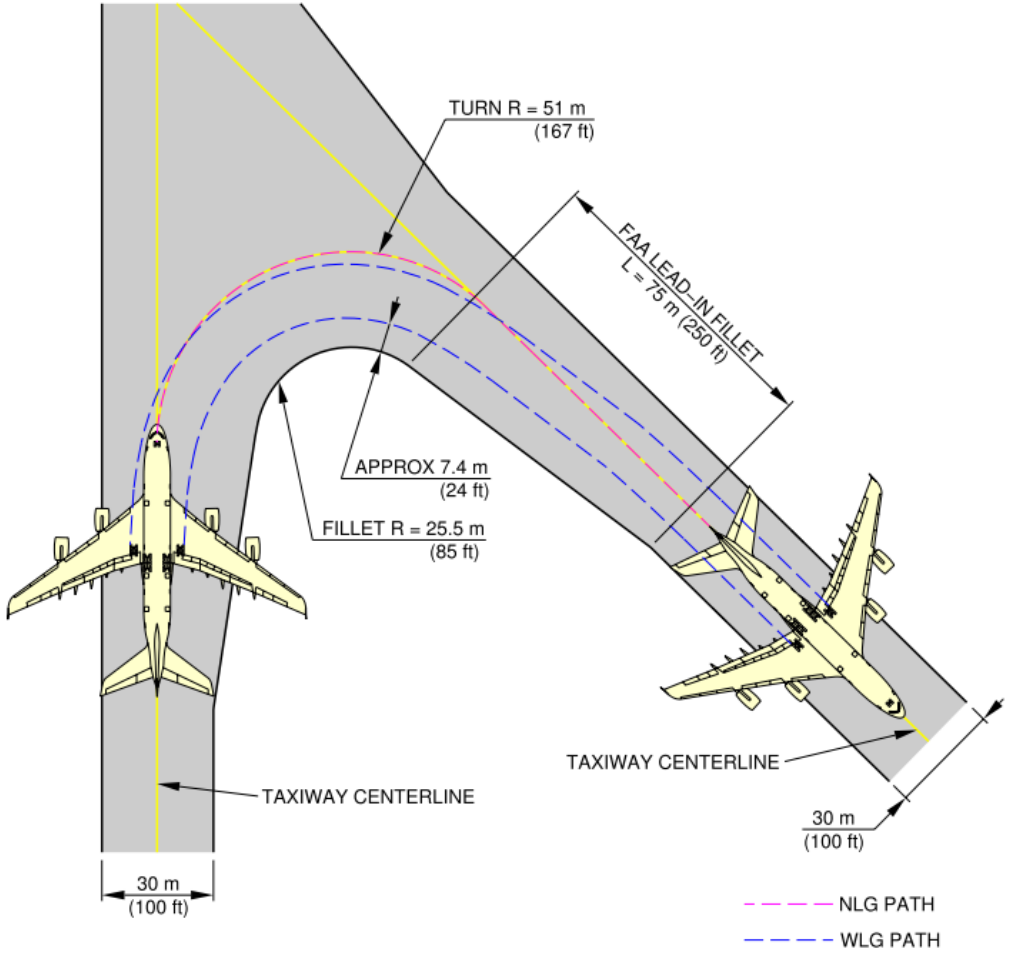
Type of Turn	Steering Angle	Effective Steering Angle		Y	A	R3	R4	R5	R6
1	70°	69.5°	m	11.08	50.91	32.66	53.76	36.52	46.01
			ft	36.3	167.0	107.2	176.4	119.8	150.9

NOTE: TURN PERFORMED WITH ASYMMETRIC THRUST AND DIFFERENTIAL BRAKING

Source: Airbus

Example: Airbus A380 (2)

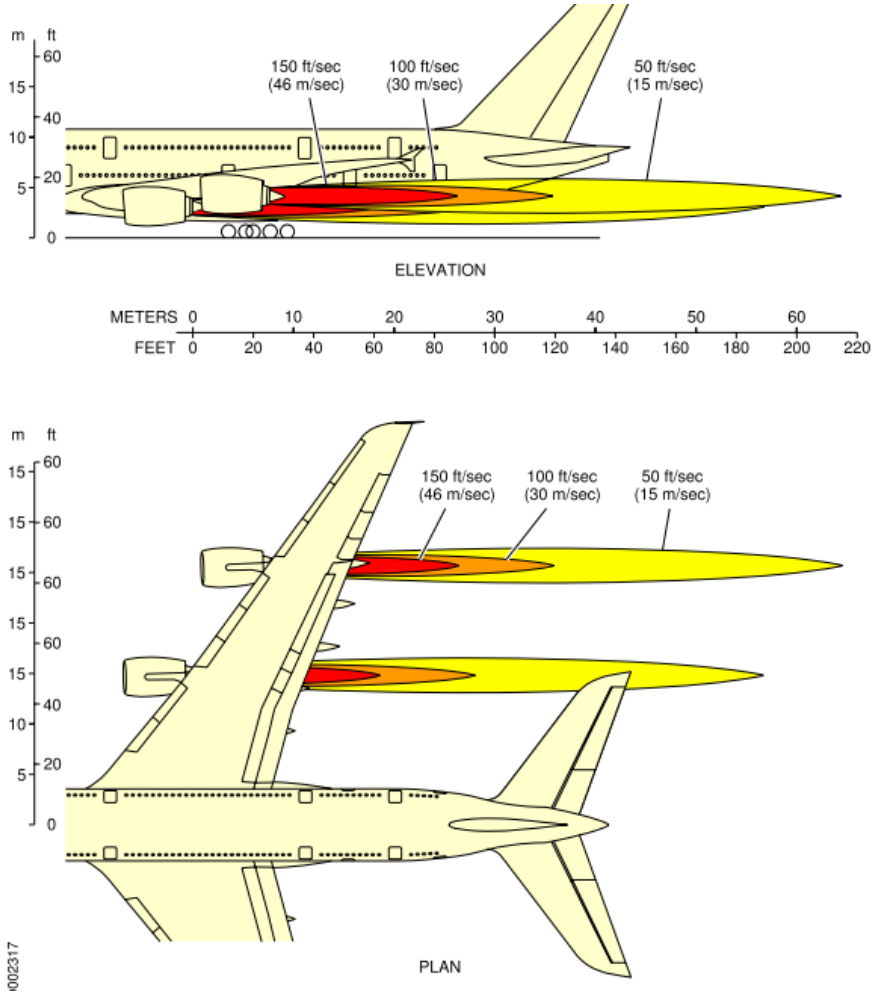
135° Turn – Taxiway to Taxiway, Cockpit Tracks Centerline Method



Source: Airbus

Example: Airbus A380 (3)

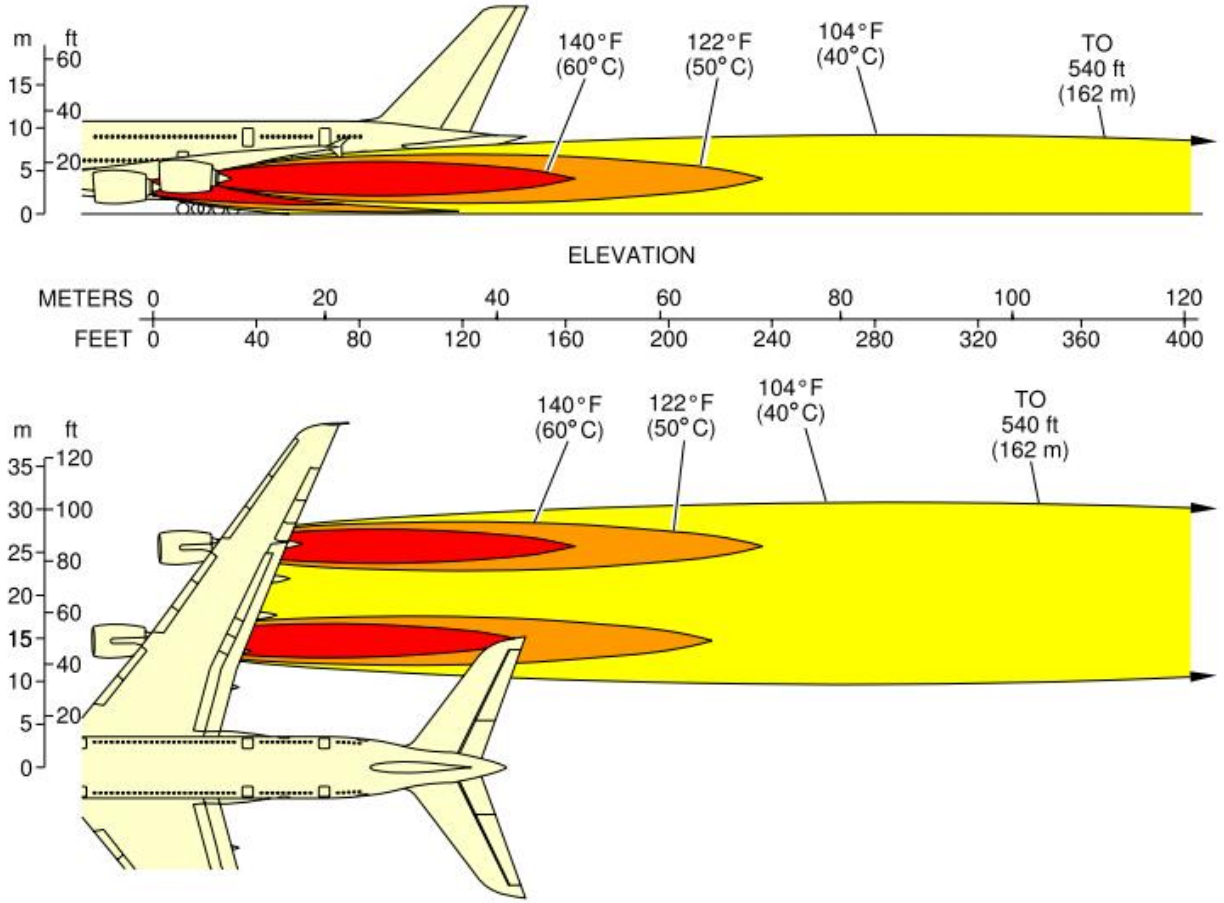
Engine Exhaust Velocities, Ground Power Idle, Trent 900 Engines



Source: Airbus

Example: Airbus A380 (4)

Engine Exhaust Temperatures, Max. Take-Off Power, Trent 900 Engines

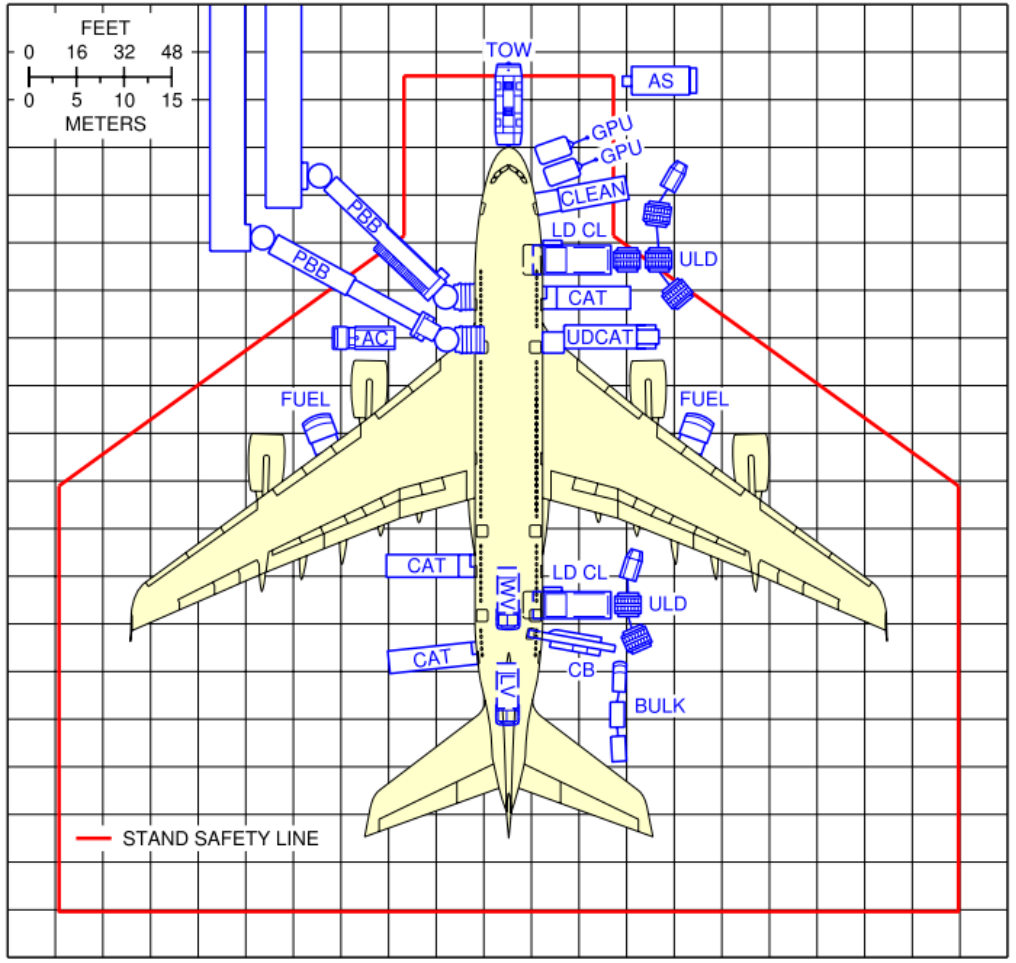


Source: Airbus

Example: Airbus A380 (5)

Terminal Servicing: Typical Ramp Layout at Gate

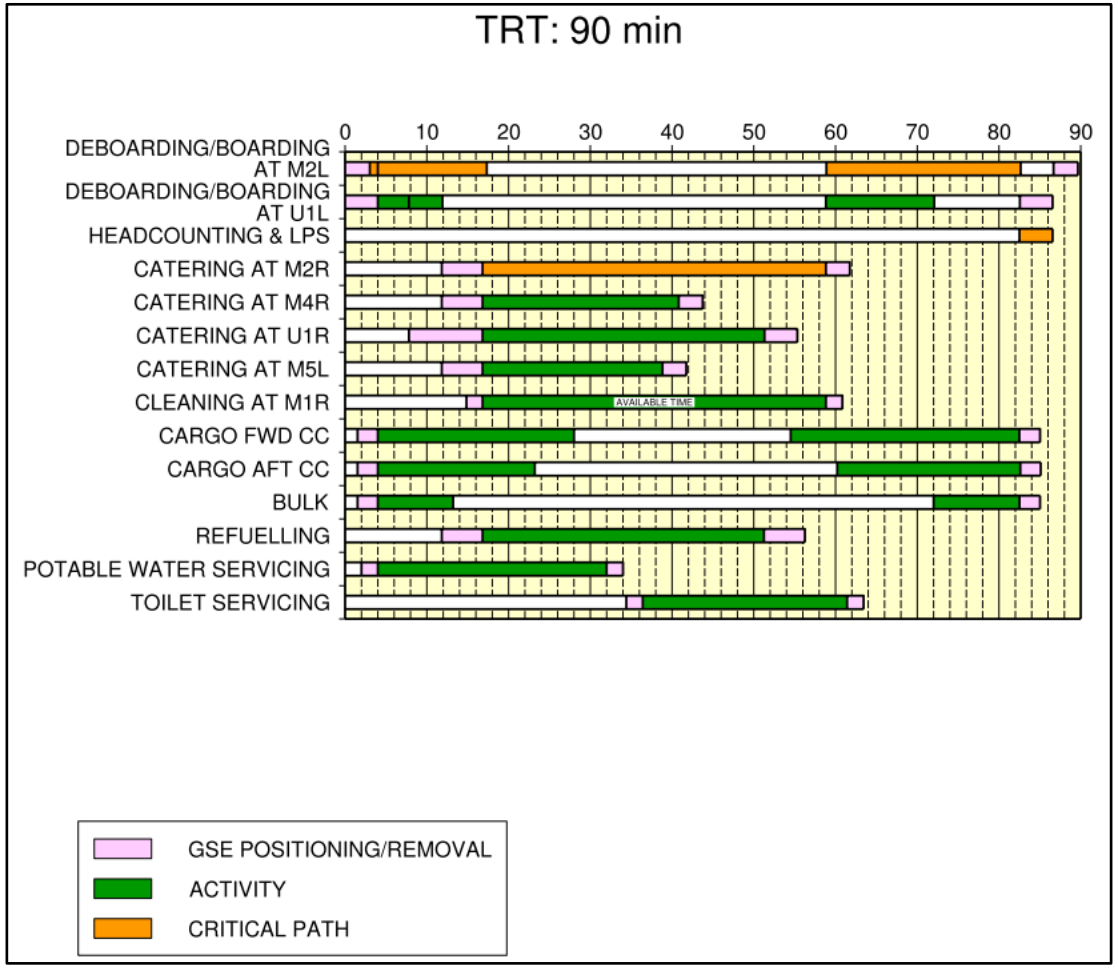
GROUND SUPPORT EQUIPMENT	
AC	AIR CONDITIONING UNIT
AS	AIR START UNIT
BULK	BULK TRAIN
CAT	CATERING TRUCK
CB	CONVEYOR BELT
CLEAN	CLEANING TRUCK
FUEL	FUEL HYDRANT DISPENSER OR TANKER
GPU	GROUND POWER UNIT
LDCL	LOWER DECK CARGO LOADER
LV	LAVATORY VEHICLE
PBB	PASSENGER BOARDING BRIDGE
PS	PASSENGER STAIRS
TOW	TOW TRACTOR
UDCAT	UPPER DECK CATERING TRUCK
ULD	ULD TRAIN
WV	POTABLE WATER VEHICLE



Source: Airbus

Example: Airbus A380 (6)

Terminal Servicing: Typical Turn-Around Time, Servicing via Main and Upper Deck with 555 Pax



Source: Airbus

The Airport Master Plan

- The airport master plan is a **development plan** for the airport facilities that is supposed to ensure a **well-coordinated and demand-oriented expansion** of the airport.

- The master plan is supposed to describe
  - where the **future development and expansion projects** will take place and
  - what **maximum potential** they may reach
  - considering a certain **time horizon**.

- The major goal of the master plan is to define the **balance of capacity**, i.e., **an economically well-balanced equilibrium among the primary traffic handling facilities**

- Further goals of the master plan are to show future perspectives (> *marketing*), to provide internal and external information, and to direct the general planning of the airport infrastructure and its financing.

Source: Gaffal (2014)

Balance of Capacity







Regional Connections
City Connections

Car Access
Car Parking

Terminals
(public, non-public)

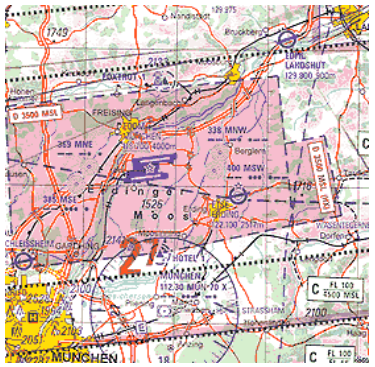
Apron

Runway

Airspace



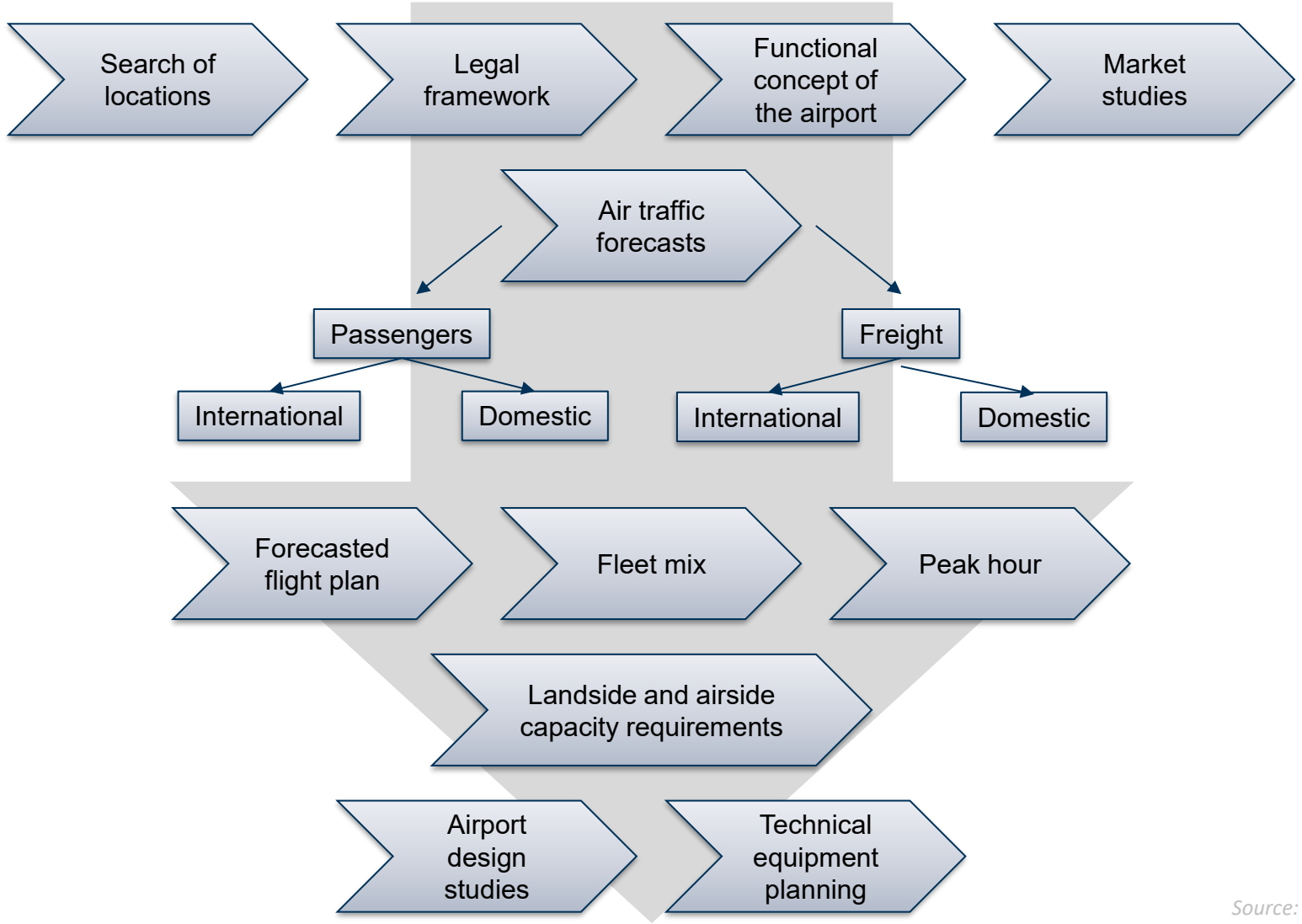




The capacity of the entire airport system is determined by the element with the smallest capacity.

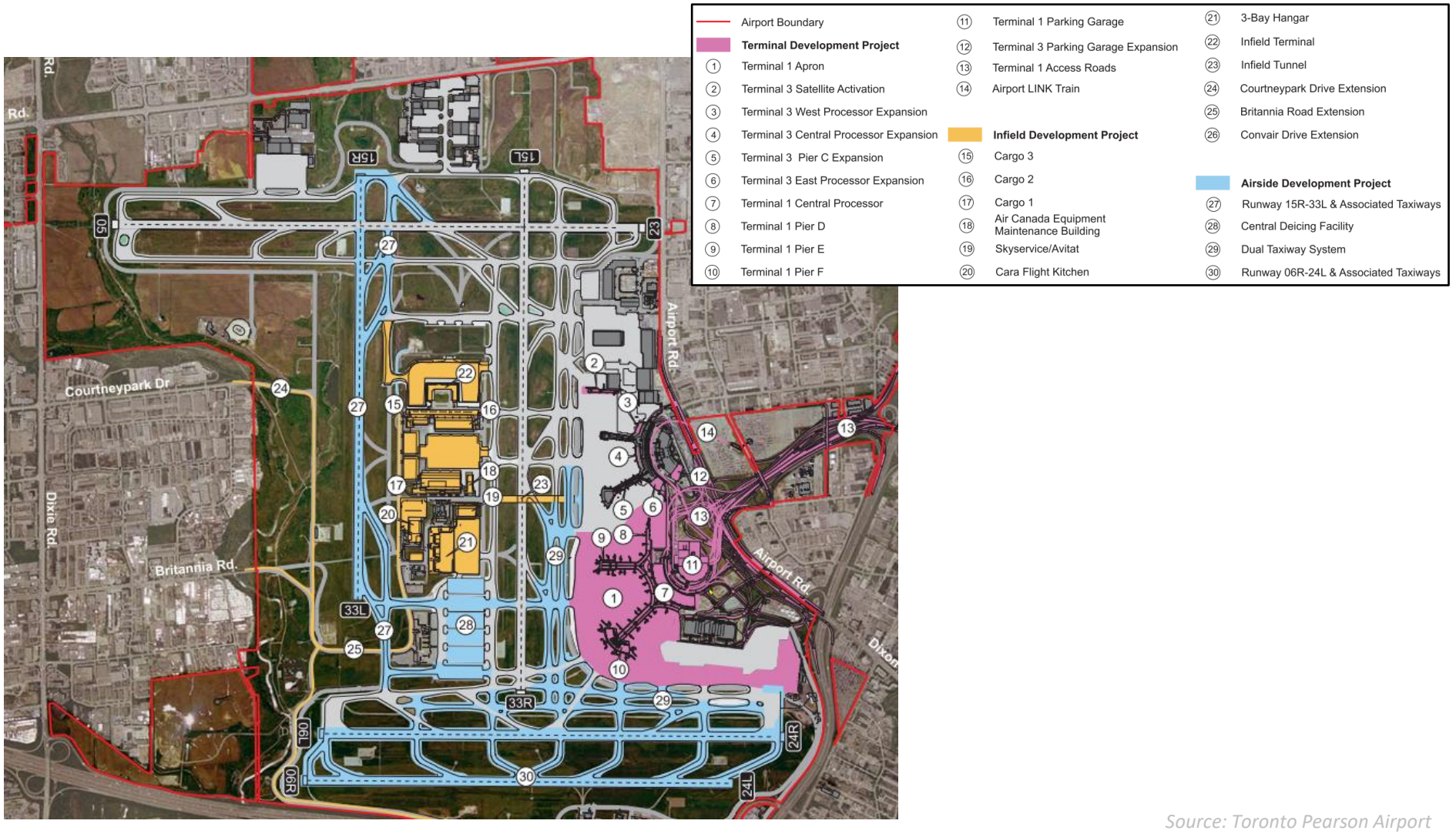
Source: Gaffal (2014)

Approach to Airport Planning



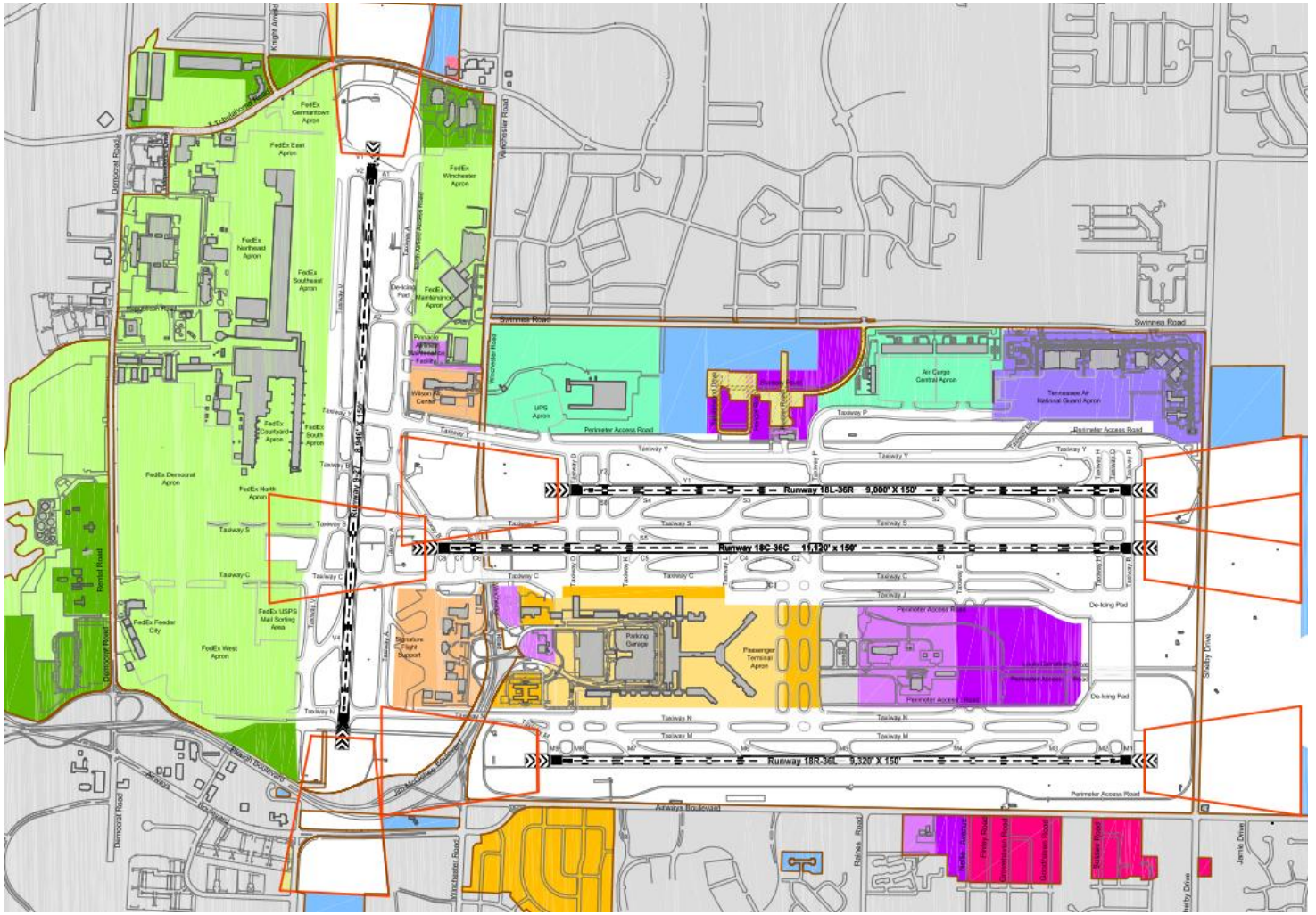
Source: Gaffal (2014)

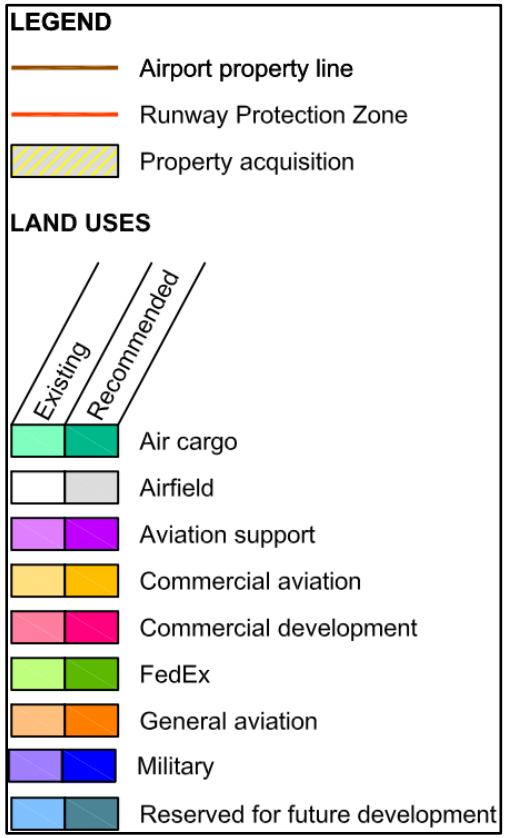
Airport Master Plan: Toronto Pearson Intl. Airport



Source: Toronto Pearson Airport

Airport Master Plan: Memphis Intl. Airport





Source: Memphis International Airport

5. Airport Operations

5.3 Airport Capacity



Airside Capacity (1)

The most important airport elements determining the airport airside capacity are **the runways, the taxiways, and the aprons**.

Runway Capacity ("Maximum throughput capacity"):

"…defined as the expected ("average") number of movements (landings and takeoffs) that can be performed per unit of time, typically 1 hour, in the presence of continuous demand and without violating air traffic control separation requirements."

The runway systems of the world's busiest airports belong to the principal **"bottlenecks"** (limiting factors) in many regions of the world.

Taxiway Capacity:

- At major airports, the taxiway capacity almost always exceeds the runway capacity by considerable margin.
- Some exceptions may exist at older airports with limited land area or inadequate taxiway systems.
- Taxiway capacity problems are airport-specific and must be resolved in the context of local conditions.

Source: Odoni (2009)

Airside Capacity (2)

Apron Capacity:

Static Capacity:

…refers to the number of aircraft stands available at the airport, i.e., the maximum number of aircraft that can be stationed on the apron simultaneously.

…stands may be designed for specific aircraft types only.

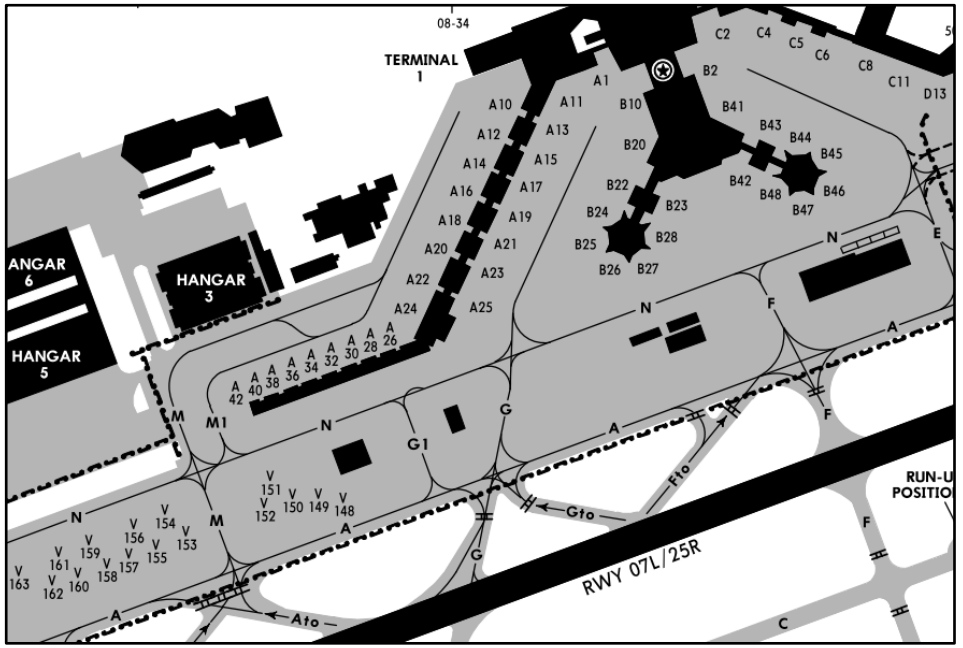
Dynamic Capacity:

…indicates the number of aircraft that can be served at the apron per unit of time, typically per hour, and is the more important measure for operating and planning purposes.

…takes into account the stand occupancy times: for a given static capacity, short stand occupancies result in a high dynamic capacity and vice versa.

The most critical question is whether the dynamic capacity is consistently greater than the capacity of the runway system, so that the apron does not constrain runway operations.

Apron at Frankfurt Airport



Source: Odoni (2009), Jeppesen

Determining the Runway Capacity (1)

The capacity of a single runway is largely determined by **the minimum separation requirements** between consecutive aircraft for same-runway operations.

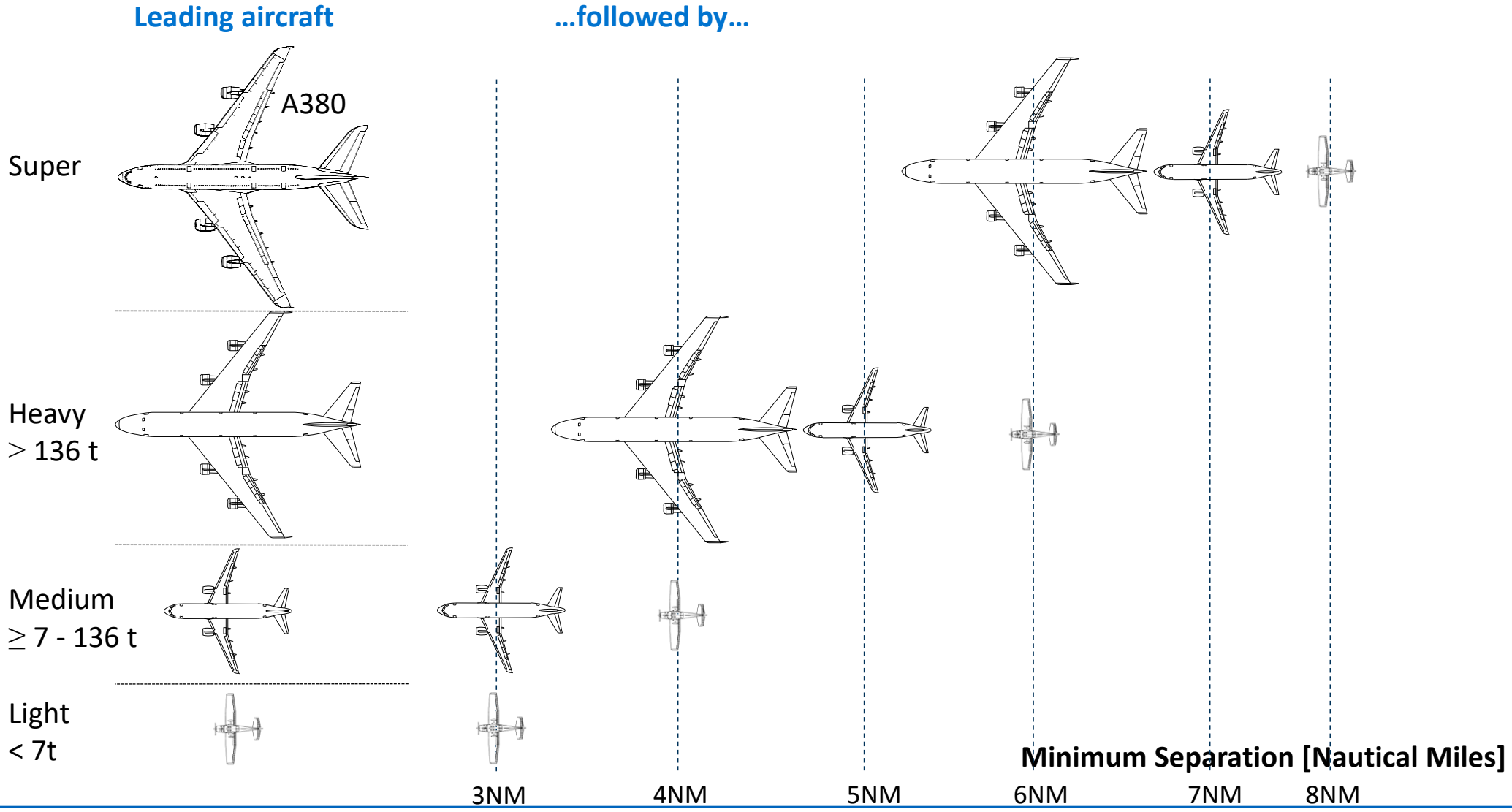
The minimum separation requirements are specified for every possible **pair of aircraft classes** (e.g., Heavy-Class Aircraft followed by Light-Class Aircraft) and for every possible **sequence of movements** (e.g., departure followed by arrival).



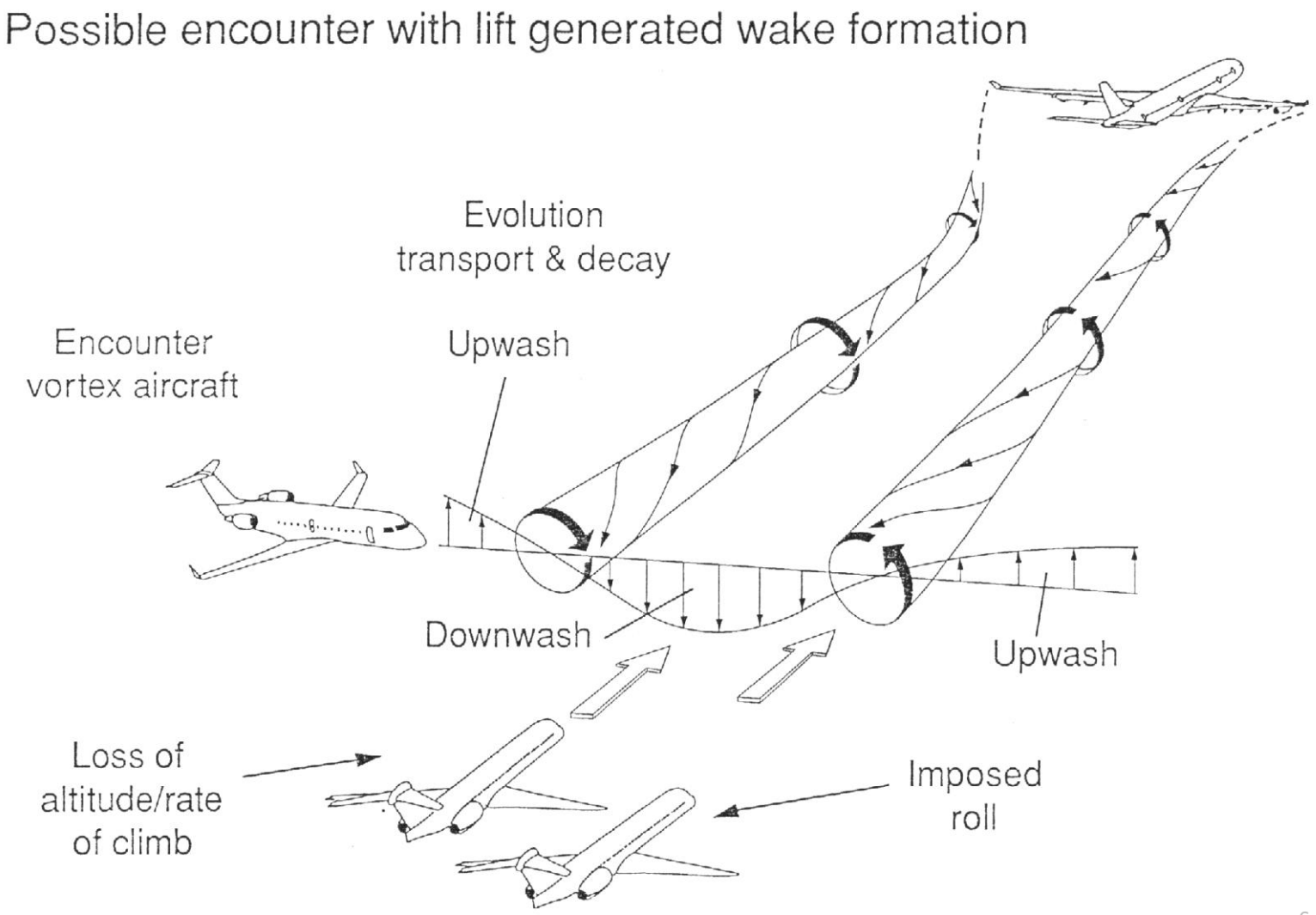
ICAO Aircraft Classification (Wake Turbulence Categories)	
Super (J)	A380 and An-225 only
Heavy (H)	MTOW > 136 tons
Medium (M)	7 tons <= MTOW <= 136 tons
Light (L)	MTOW < 7 tons

FAA Aircraft Classification (Wake Turbulences Categories)	
Super	A380 an An-225 only
Heavy	MTOW > 136 tons
Boeing 757	MTOW at the borderline btw. L and H Classes: 757 constitutes an aircraft class by itself due to its severe wake vortices.
Large	18.6 tons <= MTOW <= 136 tons
Small	MTOW < 18.6 tons

Aircraft Separation Requirements (ICAO)



Wake Turbulence



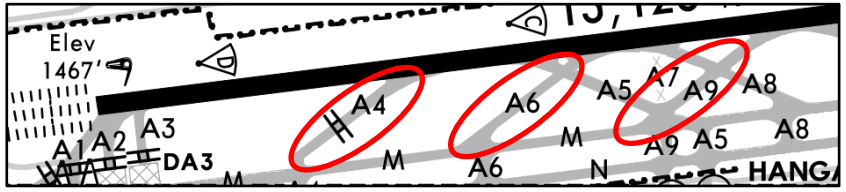
Source: Airbus

Determining the Runway Capacity (2)

Further factors influence the runway capacity:

- **Aircraft performance characteristics:** e.g., speeds on final approach influencing the runway occupancy time and the time separation between consecutive landings

- **Geometric characteristics of the runway:** e.g., availability of well-spaced high-speed exits reducing runway occupancy time on landing



- **Aircraft fleet mix:** a homogeneous mix of aircraft will result in higher capacities than when the runway is used by a diverse mix

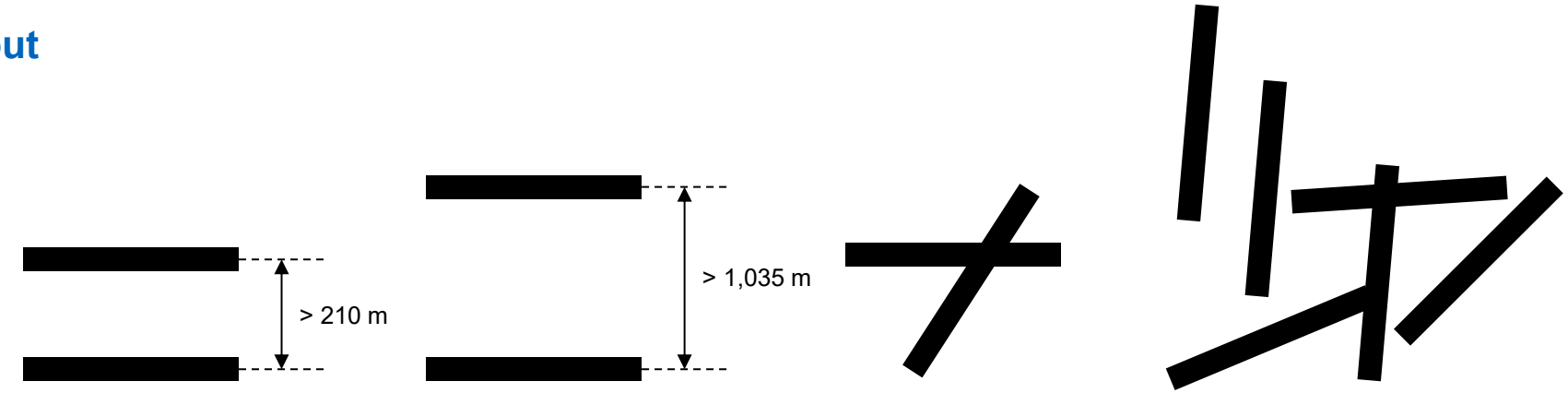
- **Types of movements** (arrivals only, departures only, or mixed operations)

Source: Odoni (2009)

Determining the Runway Capacity (3)

When a **multi-runway system** is being employed, two important factors influence the overall capacity of the runway system:

- **Geometric layout**



Closed Parallel Runway System:
A/C OPS mutually dependent
Example: Frankfurt

Open Parallel Runway System:
A/C OPS mutually independent
Example: Munich

Crossing Runway System
Example: Hamburg

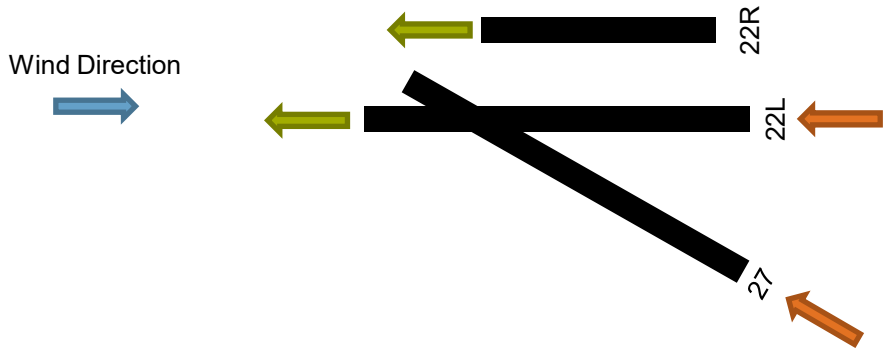
Complex Runway System
Example: Amsterdam

- **Degree of dependence between operations on different runways:** the extent to which operations on a runway may constrain operations on one or more other active runways. In the best case, movements on two different simultaneously active runways are mutually independent.

Source: Odoni (2009)

Example: Boston Logan Airport (BOS)

BOS is a multi-runway airport featuring alternative **runway configurations**, i.e., sets of simultaneously active runways with which the airport may operate at different times.



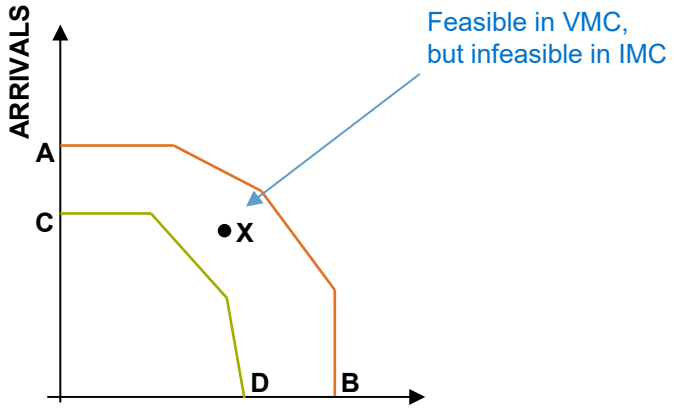
In the configuration shown above, three of the five existing runways are simultaneously active, with one runway used solely for arrivals, another solely for departures, and the third for mixed operations.

The use of the available runway configurations depends on **weather and/or noise conditions**. It is desirable to have aircraft land and take off **into the surface wind**. In calm conditions, the choice of configuration is based on such criteria as **maximizing capacity, minimizing noise impacts on neighboring communities, or retaining the opportunity to easily switch to some other favorable configuration under a forecast weather change.**

Source: Odoni (2009)

Runway Configuration Capacity Envelopes (RCCEs)

The overall airport performance can be represented as a **Pareto frontier of achievable arrival and departure rates**, called RCCEs. They provide a convenient way for displaying the capacity for any given runway configuration.

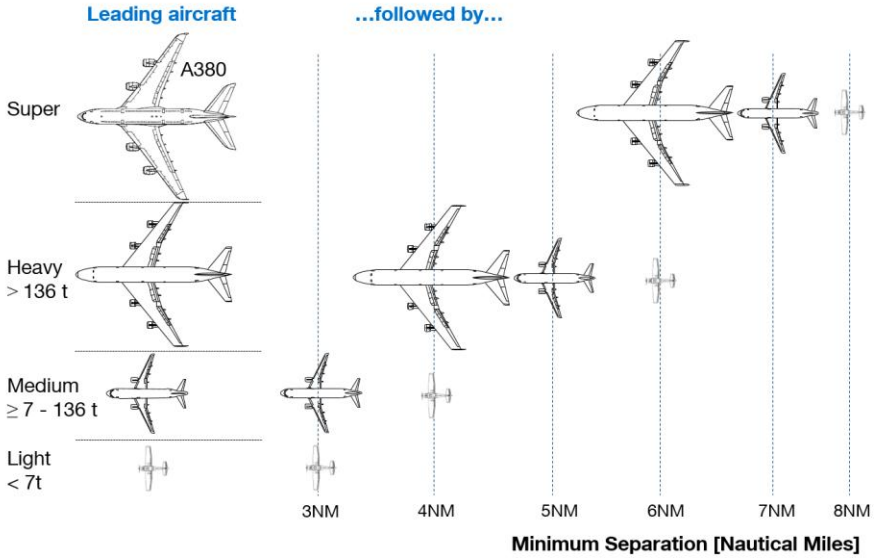


- The horizontal and vertical axes indicate the number of departures and arrivals, respectively, per unit of time (typically per hour or per 15 minutes).
- Line A-B: RCCE for a specific configuration under Visual Meteorological Conditions (VMC)
- Line C-D: RCCE for a specific configuration under Instrument Meteorological Conditions (IMC)
- Any point **inside** an RCCE corresponds to a **feasible** combination of departures and arrivals.
- Any point **outside** is **infeasible**.

Exercise: Airport Capacity (1)



Picture source: airliners.de



How many Airbus A320 can land at an airport with one runway in one hour?

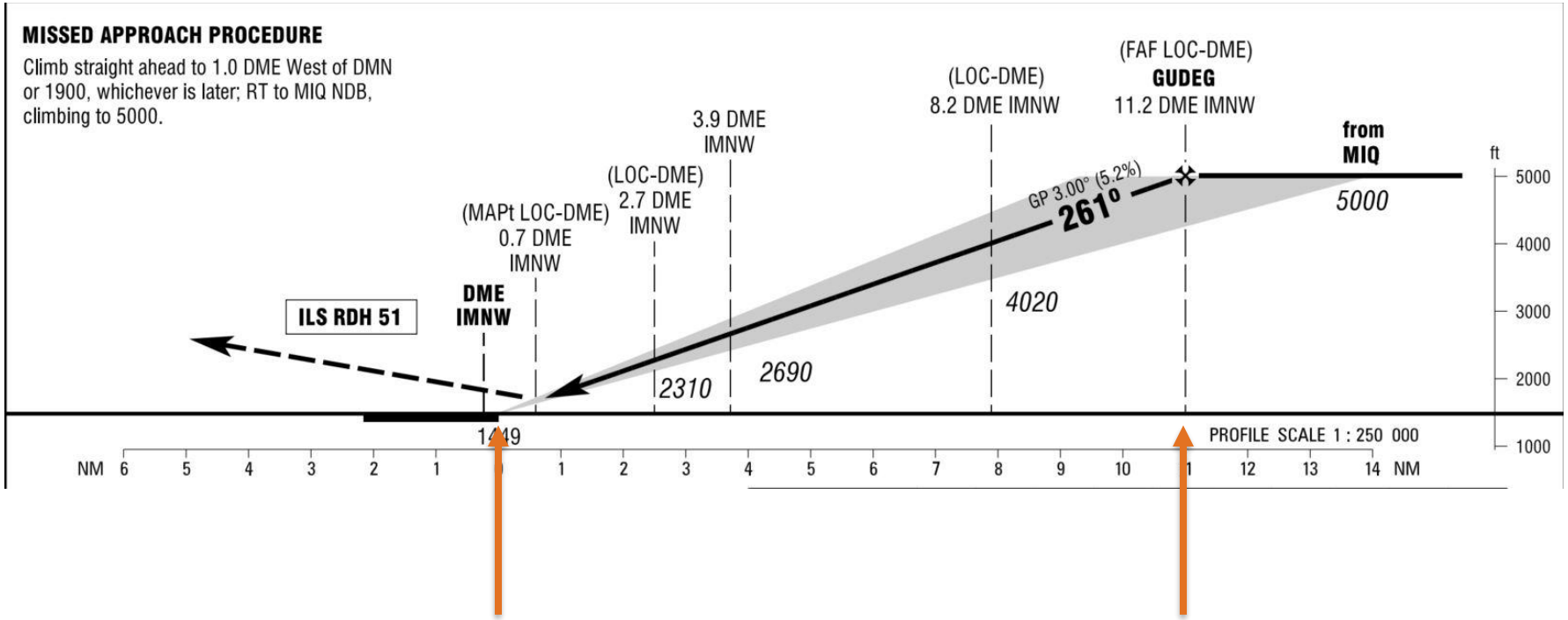
Distance separation required between A320-A320: $\boldsymbol{s_{A320-A320} = 3\ \mathrm{NM}}$
Approach **speed** Airbus A320 $\boldsymbol{v_a = 137\ KCAS}$
Space per hour: $KCAS = \frac{NM}{h} \rightarrow 137\ KCAS = 137\ \frac{NM}{h}$
Runway capacities for an Airbus A320:

$$C_{max,A320} = \frac{137\ \frac{NM}{h}}{s_{A320-A320}} + 1 = 46\ \frac{1}{h}$$

46 Airbus A320 can land on one runway (**Explanation** $\ldots + \mathbf{1}$: The first aircraft can land at t = 0 s)

Exercise: Airport Capacity (2) - Different Approach Speeds

Problem: Separation minima have to be ensured for the whole approach path



Closing Case – critical point: runway threshold
Following aircraft faster than preceding aircraft
e.g. **B737 ($v_A = 139\ KCAS$) - B747 ($v_A = 157\ KCAS$)**
Separation at **runway threshold**: $3\ NM$
Separation at final approach fix – FAF: $4.4\ NM$

Opening Case – critical point: final approach fix – FAF
Following aircraft slower than preceding aircraft
e.g. **B747 ($v_A = 157\ KCAS$) - B737 ($v_A = 139\ KCAS$)**
Separation at **runway threshold**: $6.3\ NM$
Separation at final approach fix – FAF: $5NM$

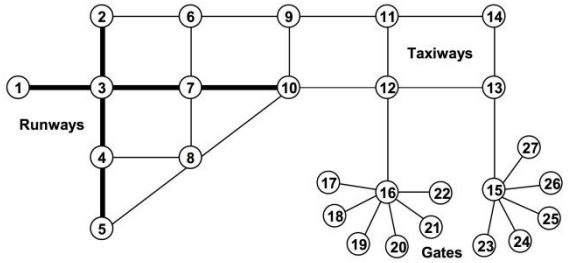
Picture source: opennav.com

Excursus: Simulation of capacity-relevant air traffic

Software: Simmod Plus! – basic principles

- discrete-event simulation software by ATAC Corporation
- used to model aircraft operation on airport ground and in the surrounding airspace.
- Both the ground infrastructure and airspace system are modeled as a series of nodes which are connected by links.

The strength of the software lies in the calculation of flight delay and the detection of the part of the system in which it occurs. This facilitates the evaluation of changes in airport layout (e.g. a new runway or gates) or future traffic demand and their impact on efficiency and capacity.



Source: Atac 2015: Simmod Pro! Reference Manual Documentation – Section 1 figure 6-1

